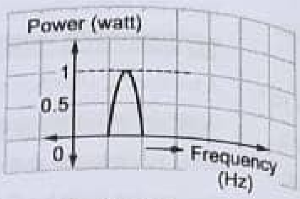
4.1 INTRODUCTION

A. Narrowband Signal

What is narrowband signal? *(For reading purpose only...)*

The signal which utilizes narrower band of frequencies in the communication channels is known as the 'narrowband signal'. Typically narrowband signals are used for carrying signals like audio spectrums of limited bandwidth. It makes use of lesser bandwidth. *Refer Fig. 4.1.1.*

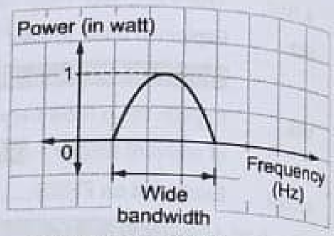


(1F1) Fig. 4.1.1 : Narrowband

B. Wideband Signal

What is a wideband signal? *(For reading purpose only...)*

The signal which makes use of wide band of frequencies in the communication channels is known as 'wideband signal'. *Refer Fig. 4.1.2.*



(1F2)Fig. 4.1.2 : Wideband spectrum

C. Comparison between Narrowband and Wideband Signal *(For reading purpose only ...)*

Sr. no.	Parameter	Narrowband signal	Wideband signal
1	Definition	The signal which utilizes narrower band of frequencies in the communication channels is known as the narrowband signal.	The signal which makes use of wide band of frequencies in the communication channels is known as wideband signal.
2	Advantages	Better range of reception as design of narrower filters is simple. It can be used when slow data streams are to be transmitted	As wide bandwidth is used, faster communication achieved. It allows reduction of narrowband noise It provides high degree of security of data
3	Disadvantages	Slow communication speed as narrower bandwidth is used. Noise can affect the signal easily as less security measures are used.	Better linearity of filters is needed. Expensive systems.
4	Modulation schemes used	AM, FM, SSB, BPSK, etc.	OFDM, GMSK, etc.
5	Applications	AM/FM radio, satellite downlinks, GPS signals, etc.	Wi-Fi, 4G LTE, OFDM, etc.

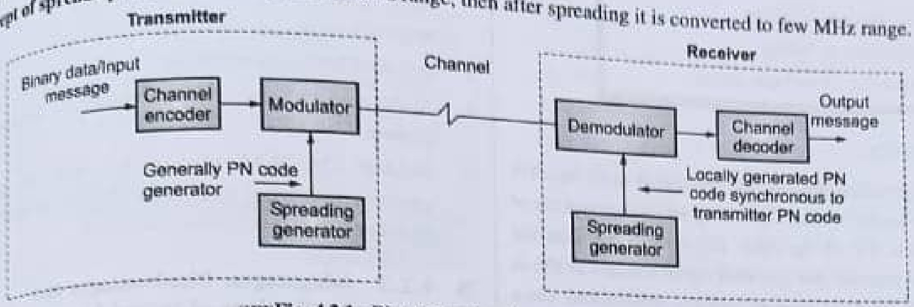
4.2 CONCEPT OF SPREADING THE SPECTRUM

UQ. Draw the block diagram of spread spectrum digital communication and explain the various blocks.
SPPU - (E&TC-Sem. 5) Dec. 17, 8 Marks

Ans. :

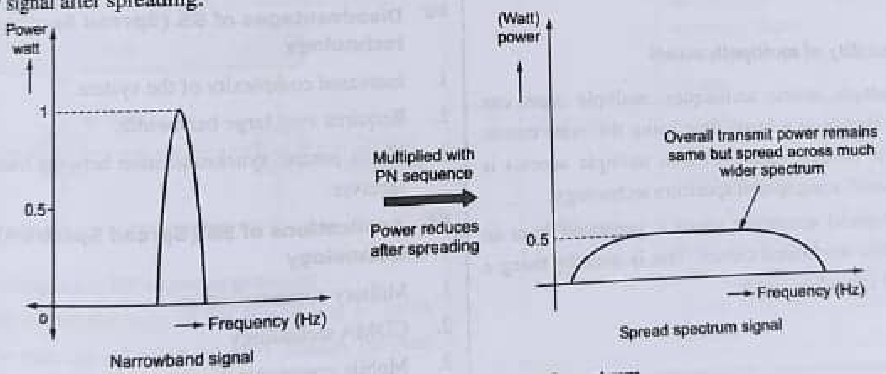
Definition : The process in which narrowband signal is converted to wideband signal is called as 'spreading of spectrum's.

Concept of spread spectrum: If the signal is in kHz range, then after spreading it is converted to few MHz range.



(1F3)**Fig. 4.2.1 : Block diagram of spread spectrum system**

1. *Refer Fig. 4.2.1.* It shows block diagram of spread spectrum system. Binary data is given as the input to the channel encoder. This binary data is narrowband signal. Generally channel encoder adds **redundant bits** to the input binary data bits and provides it to the modulator.
2. The modulator block actually comprises of multiplier and digital CW modulators. Multiplier has the other input from PN sequence or spreading code generator. This spreading code is wideband noise like signal. When the binary data is multiplied (**modulo - 2 added**) with the PN code, a wideband signal is obtained at the output.
3. This is **spreaded binary code.** This is then modulated using MPSK or MFSK modulators and then transmitted over channel. At the receiver, the signal is first demodulated by coherent or non coherent detection methods. Once the spreading code is detected back, it is again multiplied with the locally generated PN code. Then the original binary data is recovered back.
4. The power of the narrowband signal is reduced to almost half or even less than that, after spreading the signal. This is shown in Fig. 4.2.2. ***Refer Fig. 4.2.2.*** But the overall transmit power remains same even after spreading the signal. This power gets distributed over wider spectrum. In earlier digital communication, entire power was concentrated in tight bandwidth. This is not done in spread signal. For example, **6 MHz, 0.3 mW** signal may get converted into **22 MHz, 0.1mW** signal after spreading.



(1F4)**Fig. 4.2.2 : Concept of spread spectrum**

Unit IV End Sem.

4.2.1 Need of Spreading the Spectrum

UQ. What is need of spread spectrum modulation technique ?

SPPU - (E&TC-Sem. 5) May 16, 4 Marks

Ans. :

1. Security
2. Resistance to jamming
3. Resistance to fading
4. Capability of multiple access

1. Security

In the spreading process, the sequential noise like PN (pseudo code) is used. Spreading allows encryption of the signal. To all the other receivers, other than the intended receiver, the spreaded signal appears as if it is noise. It has been used in military applications since 1940, as security is the major necessity. Later, it was used in commercial mobile applications.

2. Resistance to jamming

In spread spectrum signals, the jamming power is spread all over the spectrum. As the spectrum is wide enough, the overall effect of jamming is very much less than the background noise. On the contrary, in narrowband signals, as the bandwidth is less, the effect of jamming power is significant. Jamming severely degrades the quality of signal.

3. Resistance to fading

Wide bandwidth signal supports some sort of frequency diversity. Therefore, signal faces multipath fading very rarely.

4. Capability of multipath access

- With multiple access techniques, multiple users can transmit signals at a same time using the same carrier frequency. CDMA (code division multiple access) is implemented using spread spectrum technology.
- The SS (spread spectrum) signal is generated from an already data modulated carrier. This is done by using a wideband spreading signal.

4.2.2 Conditions for Spread Spectrum Signal

GQ. State conditions for spread spectrum signal.

Ans. :

1. The transmission bandwidth should be much greater than the minimum signal bandwidth required for transmission.
2. For the purpose of spreading, codes should be used and these codes need to be independent of the data sequence.
3. It is done before transmission.
4. At the receiver, same code is necessary for dispreading purpose.
5. Standard FM, PCM, etc. are not called as spread spectrum modulation schemes, as they do not satisfy all these conditions.

4.2.3 Advantages, Disadvantages and Applications of Spread Spectrum Technology

GQ. State advantages, disadvantages and applications of spread spectrum technology.

Ans.:

Advantages of SS (Spread Spectrum) technology

1. It has relatively high security.
2. SS system has better noise immunity by resistance to narrowband jamming of the signal.
3. This system supports better performance in multipath fading environment.
4. SS supports multiple access schemes like CDMA.

Disadvantages of SS (Spread Spectrum) technology

1. Increased complexity of the system.
2. Requires very large bandwidth.
3. Needs precise synchronization between transmitter and receiver.

Applications of SS (Spread Spectrum) technology

1. Military communication
2. CDMA technology
3. Mobile communication

4.2.4 Classification of Spread Spectrum Modulation Techniques

There are four types of spread spectrum techniques depending on the type of insertion of PN code

1. DSSS : Direct Sequence Spread Spectrum
2. FHSS : Frequency Hopped Spread Spectrum
3. THSS : Time Hopping Spread Spectrum
4. Hybrid Spread Spectrum

4.3 PSEUDO NOISE SEQUENCE (PN) CODE

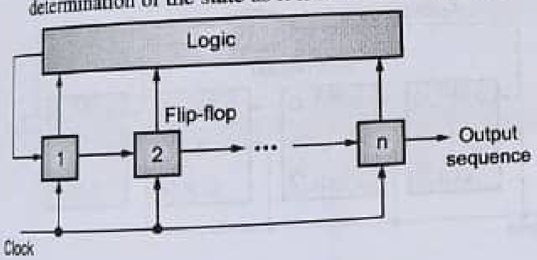
GQ. Explain PN sequences generator with neat block diagram.

UQ. What is PN sequence ? **SPPU - (E&TC-Sem. 5) May 15, Dec. 16, May 17, 2 Marks**

Ans. :

Definition : PN' code is the deterministic and periodic binary sequence generated by feedback logic circuit. It provides noise like waveform at the output.

1. *Refer Fig. 4.3.1.* It shows the general block diagram of PN sequence generator. The design of PN sequence generator (**feedback shift register**) consists of 'n' number of flip-flops and the feedback logic circuit. It forms the multiloop feedback circuit. A single timing clock is given to the generator.
2. It is the shift register. When the clock pulse is applied, the output of the flip-flop is shifted to the input of the next flip-flop and next state is obtained. The output of PN sequence is taken at the final stage flip-flop.
3. At every state, the feedback logic also contributes in determination of the state as it is in feedback circuitry.



(1F5)**Fig. 4.3.1: PN sequence generator**

4. Let $m_j(k)$ denote the state of the flip-flop at k^{th} clock pulse. The state can either be '0' or '1'.Hence, the state of the shift register after k^{th} clock pulse can be given as a set of $\{m_1(k), m_2(k), m_3(k), \ldots\ldots m_n(k)\}$;

where, $k \geq 0$

5. Let for initial state, $k = 0$, Therefore,

$$m_j(k+1) = m_{j-1}(k), \ \{k \geq 0 \ ; \ \{1 \leq j \leq n$$

6. If 'n' number of the flip-flop are used, then the number of states in PN sequence is equal to 2^n.If the feedback logic consists of modulo-2 adders, then the '0' state, i.e., output state of all the flip-flops is zero, is not allowed.
7. This is because, if the initial state is kept zero and as there are modulo-2 adders in the feedback logic, then the shift register will continue with the same state and will not advance to next state. This is deadlock situation. Hence, the number of states in the PN generator $= 2^{n-1}$. (One state is omitted which is all zero state).Maximal length sequence or m sequence: The PN sequence is known as maximal length sequence when the states are equal to 2^{n-1}.

4.3.1 Comparison between Pseudo Random Signal and the Random Signal

GQ. What is the difference between the Pseudo random signal and the Random signal?

Ans. :

Table 4.3.1 : Comparison between Pseudo random signal and the random signal

Sr. No.	Parameters	Pseudo random signal	Random signal
1.	Prediction	It is deterministic and periodic sequence; hence, can be predicted exactly.	It cannot be predicted exactly.
2.	Analysis	It can be analysed mathematically.	It needs statistical analysis.
3.	Nature of signal	It is known to both transmitter and receiver in spread spectrum systems.	It is totally random, so cannot be generated.

Unit IV

4.3.2 PN sequence as Pseudo Random

Why PN sequence is known as pseudo random noise like signal?

GQ. Why PN sequence is known as pseudo random noise like signal?

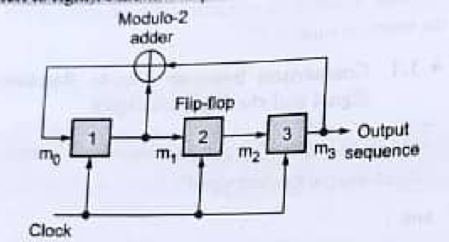
Ans. :

Even though the signal is deterministic, it appears as if it is white noise or a truly random signal to the unauthorized receiver. Hence, it is called as noise like signal.

4.3.3 Examples on PN Sequence

UEx. 4.3.1 SPPU - (E & TC-Sem. 5) May 17, 6 Marks

Design a three stage feedback shift register with proper taps to generate N = 7 PN sequence. Draw the generator block and if the initial state of shift register is 100 (from left to right). Find the output sequence.



(1F6)Fig. P. 4.3.1 : PN sequence generator with n=3

Soln. :

1. Number of flip-flop = 3

   Therefore, n = 3,

   Hence, number of states $N = 2^n - 1 = 2^3 - 1 = 8 - 1 = 7$

2. The modulo-2 adder can be simple EX-OR logic gate.

Truth table of EX-OR gate

Input		Output (Y)
A	B	
0	0	0
0	1	1
1	0	1
1	1	0

3. Now looking at the diagram, it is clear that output of modulo-2 adder = $m_1 \oplus m_2$. Hence, the truth table considering state 100 as initial state is shown in Table P. 4.3.1.

Table P. 4.3.1 : Truth table of PN sequence generator

Clock	m_1	m_2	m_3
1	1	0	0
2	1	1	0
3	1	1	1
4	0	1	1
5	1	0	1
6	0	1	0
7	0	0	1
8	1	0	0

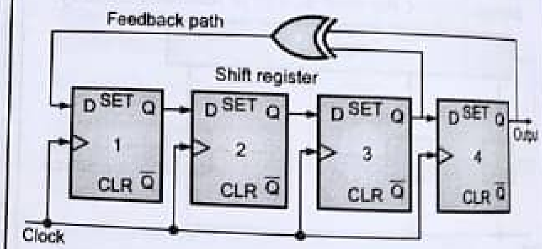
4. The output sequence (output of the last stage flip-flop) of the shift register is therefore, 0011101.... The sequence will repeat after 7^{th} state as shown in Table P. 4.3.1.

Note : Any of the six states can be taken as initial state.

UEx. 4.3.2 SPPU - (E & TC-Sem. 5) May 16, 9 Marks

Generate the PN sequence for transmitting message through FHSS system. The period of PN sequence is $2^4 - 1 = 15$. The initial content of shift register is assumed to be 1100. Draw PN sequence generator.

Soln. :



(1F23)Fig. P. 4.3.2 : PN sequence generator

Clock pulse	Q_4	Q_3	Q_2	Q_1	$Q_3 \oplus Q_4$	PN Code
1	1	1	0	0	0	1
2	1	0	0	0	1	1
3	0	0	0	1	0	0
4	0	0	1	0	0	0
5	0	1	0	0	1	0
6	1	0	0	1	1	1
7	0	0	1	1	0	0
8	0	1	1	0	1	0
9	1	1	0	1	0	1
10	1	0	1	0	1	1
11	0	1	0	1	1	0
12	1	0	1	1	1	1
13	0	1	1	1	1	0
14	1	1	1	1	0	1
15	1	1	1	0	0	1
16	1	1	0	0	0	1

Initial state

(1F24)Fig. P. 4.3.2(a)

4.3.4 Properties of Maximal Length Sequence / PN Sequence

UQ. What are the properties of maximal length sequences? Give the graphical representation of auto correlation property of random data and a PN sequence and explain.

SPPU - (E&TC-Sem. 5) Dec. 14, 8 Marks

UQ. State the properties of PN sequence with the help of 4 stage shift register.

SPPU - (E&TC-Sem. 5) May 15, 6 Marks, Dec. 17, 6 Marks

UQ. Verify the three properties of PN sequence with the help of shift register.

SPPU - (E&TC-Sem. 5) Dec. 16, May 17, 6 Marks

☑ **Ans. :**

1. Balance property
2. Run property
3. Correlation property

1. Balance property

In each period of the PN sequence, the number of 1's is always one more than number of 0's. E.g.in the sequence in Ex. 4.3.1 the number of 1's are 4 and number of 0's are 3. Hence, number of 1's is one more than number of 0's.

2. Run length property

Definition of Run : A run is defined as the string of continuous identical binary digits.

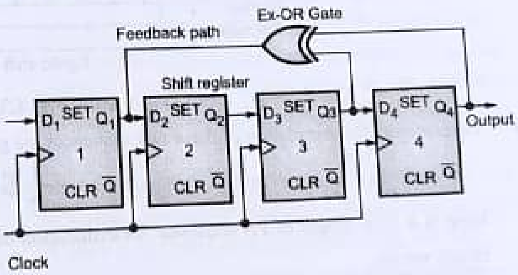
- New binary digit in the sequence changes the run. The run length is the number of binary digits in the run. The run property states that one half the runs of each type of length is equal to 1, about one fourth are of length 2, one eighth are of length 3 and so on.
- For a PN sequence generated by 'n' number of flip-flops, the total number of runs is **(N + 1)/2**, where $N = 2^n - 1$. Consider a PN code of length = 15. It means 4 flip-flops are used and 15 states are generated. This is 4 bit PN sequence generator as shown in Fig. 4.3.2.

4 bit PN sequence generator

UQ. Draw and explain 4 bit PN sequence generator and find maximum length sequence.

SPPU - (E&TC-Sem. 5) Dec. 15, 8 Marks

☑ **Ans. :**



(1F7)Fig. 4.3.2 : 4 bit PN sequence generator

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Table 4.3.2 : The states obtained

Clock	Q_1	Q_2	Q_3	Q_4	PN code
					0
1	1	0	0	0	0
2	0	1	0	0	0
3	0	0	1	0	1
4	1	0	0	1	0
5	1	1	0	0	0
6	0	1	1	0	1
7	1	0	1	1	1
8	0	1	0	1	0
9	1	0	1	0	1
10	1	1	0	1	0
11	1	1	1	0	1
12	1	1	1	1	1
13	0	1	1	1	1
14	0	0	1	1	1
15	0	0	0	1	0
16	1	0	0	0	

- The PN code obtained is **000100110101111**. In this PN code sequence,

1. Total number of runs = 8
2. One run of length = 3 = $1/8^{th}$ of runs of length 3
3. Two runs of length = 2 = $1/4^{th}$ of runs of length 2
4. Four runs of length = 1 = 1/2 runs of length 1

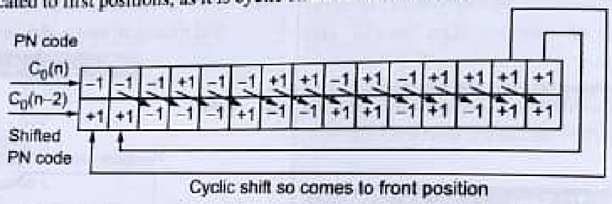
If the sequence has more number of states, then it goes on. This is known as 'the run length property of the PN sequence'.

3. Cyclic shift property / Autocorrelation Property

Consider a PN sequence shown in Fig. 4.3.2. The PN sequence is 000100110101111. We will allocate – 1 for logic '0' and + 1 for logic '1'. Hence, the sequence becomes,

PN code sequence →	0	0	0	1	0	0	1	1	0	1	0	1	1	1	1
+ 1 and – 1 level for PN codes →	– 1	– 1	– 1	+ 1	– 1	– 1	+ 1	+ 1	– 1	+ 1	– 1	+ 1	+ 1	+ 1	+1

- This code sequence is cyclically shifted in time, For example, the code shown above is shifted by two time periods and last two bits are allocated to first positions, as it is cyclic shift. Hence, we get,



(1F26) **Fig. 4.3.3 : Cyclic shift**

- Determining the correlation of these two sequences we get,

$$= \frac{1}{N}\sum C_0(n) \cdot C_0(n-2).....$$

Here N = 15 = length of PN sequence. Determination of correlation is done by taking product of $C_0(n)$ and $C_0(n-2)$. Hence we get,

PN code →	$C_0(n)$	– 1	– 1	– 1	+ 1	– 1	– 1	+ 1	+ 1	– 1	+ 1	– 1	+ 1	+ 1	+ 1	+1
Shifted PN codes →	$C_0(n-2)$	+ 1	+ 1	– 1	– 1	– 1	+ 1	– 1	– 1	+ 1	+ 1	– 1	+ 1	– 1	+ 1	+1
Product	$C_0(n)\,C_0(n-2)$	– 1	– 1	+ 1	– 1	+ 1	– 1	– 1	– 1	– 1	+ 1	+ 1	+ 1	– 1	+ 1	+1

$$\therefore \sum C_0(n)\,C_0(n-2) = -1$$

$$\therefore \text{ Correlation } R(\tau) = \frac{1}{N}\Sigma C_0(n)\, C_0(n-2)$$

$$\therefore \; R(\tau) = \frac{1}{15}\Sigma C_0(n)\, C_0(n-2) = \frac{-1}{15}$$

This is true for any shift. Hence it can be concluded that, the correlation for any shift other than zero is $\frac{-1}{N}$. For shift 0,

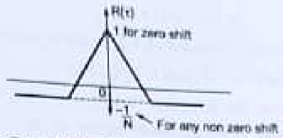
$$\text{Correlation} = \frac{1}{N}\Sigma C_0(n)\, C_0(n)$$

$$\therefore \; R(\tau) = \frac{1}{N}\Sigma C_0^2(n)$$

$$\therefore \; R(\tau) = \frac{1}{N}\Sigma 1 = \frac{N}{N}$$

$$\therefore \; R(\tau) = 1$$

$$\text{Correlation of PN sequence } \begin{cases} 1 & \text{if shift} = 0 \\ \frac{-1}{N} & \text{For any, other non-zero shift} \end{cases}$$



(1F8)Fig. 4.3.4 : Correlation diagram for PN sequence

4.4 A NOTION OF SPREAD SPECTRUM

(For reading purpose only.)

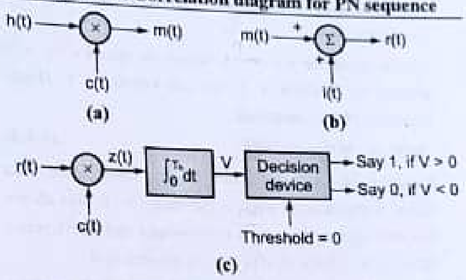
1. The spread spectrum signals provide better protection against the **narrowband jamming signals**. These jamming signals may consist of fairly powerful broadband noise or multitone waveform which may disrupt the communication signals.
2. It is provided by making the signal occupy wider bandwidth than **minimum transmission bandwidth** required. Signal is transmitted by hiding behind the wide bandwidth signal. It also gives the spreading signal an appearance of noise like signal. Therefore, it can smoothly pass through the channel without anyone listening to the signal. Hence, better security is provided to the signal.

4.4.1 Method of Spreading the Signal

Q. Explain ideal system of spread spectrum technology.

Ans. : *Refer Fig. 4.4.1.*

1. Binary data sequence (user data) used is always in NRZ (Non Return to Zero) format. Let b (t) be the binary data sequence with levels ±1 and c (t) be the PN code.
2. b (t) is the narrowband signal and c (t) is wideband signal. b (t) and c (t) are applied to the product modulator or the multiplier.



(1F9)Fig. 4.4.1 : Ideal system of spread spectrum technology (a) transmitter (b) channel (c) receiver

3. As per Fourier transform theory, the multiplication of the narrowband and wideband signals will give convolution of the spectrum of two individual signals. Hence, the product of the narrowband spectrum and the wideband spectrum is nearly equal to the wideband spectrum. Hence, it can be said as PN code spreads the signal.
4. After multiplication process, the user data or the information signal b (t) is actually chopped into number of small times increments as shown in Fig. 4.4.1(b).
5. These small increments are known as chips. Hence,

$$x(t) = b(t)\, c(t) \qquad \text{...(4.4.1)}$$

6. At the receiver, signal $r(t) = [x(t) + i(t)]$ is received.

where, i(t) = interference added

$$\text{Therefore, } r(t) = b(t)\, c(t) + i(t) \qquad \text{...(4.4.2)}$$

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7. To detect the original information signal back at the receiver, the received signal is given to demodulator circuit.

8. ***Refer Fig. 4.4.1(c).*** Demodulator consists of multiplier, integrator and the decision device as shown in Fig. 4.4.1(c). Locally generated PN code is provided to the multiplier. For generation of the exact PN code as that of transmitter, the transmitter and receiver has to maintain synchronism.

9. Hence the multiplier output is given as,

$$\mathbf{m(t) = c(t)\ r(t)}$$

Putting Equation (4.4.2) in above equation we get,

$$\mathbf{m(t) = c^2(t)\ b(t) + c(t)\ i(t)} \quad ...(4.4.3)$$

11. From the Equation (4.4.3), it is clear that information signal is multiplied by square of the wideband PN code and unwanted interference signal is multiplied by single PN signal.

c(t) can either be + 1 or − 1, hence, its square $c^2(t)$ will always be positive + 1, for all values of t. Hence Equation (4.4.3) becomes

$$\mathbf{m(t) = b(t) + c(t)\ i(t)} \quad ...(4.4.4)$$

Equation (4.4.4) shows that binary information data signal is reproduced back at the receiver. It also shows that data signal received is narrowband and interference signal is wideband as **c(t)** term is present in it.

13. Hence, this signal is now applied to LPF (Low Pass Filter). LPF will pass only narrowband signal and wideband interference is automatically filtered out.

14. This filtering action is performed by the integrator. Integration is carried out for one bit duration. Integrator's output is given to the decision device. If the incoming signal sample value is greater than the threshold set at the decision device, then **logic '1'** is detected otherwise **logic '0'**.

4.5 DIRECT SEQUENCE SPREAD SPECTRUM(DSSS)

UQ. Draw block diagram of DS- PSK transmitter.

SPPU - (E&TC-Sem. 5) Dec. 14, 2 Marks

UQ. Draw the block diagram of DSSS system transmitter and receiver. Write functional names inside the block and input signal for each block and explain.

SPPU - (E&TC-Sem. 5) May 17, 6 Marks

UQ. Explain DS-BPSK transmitter and receiver with suitable block diagram.

SPPU - (E&TC-Sem. 5) May 17, 3 Marks

UQ. Explain DSSS transmitter and receiver with neat waveforms.

SPPU - (E&TC-Sem. 5) May 18, 6 Marks

☑ **Ans. :**

1. ***Refer Fig. 4.5.1 (a).*** In DSSS, signal undergoes two stages of modulation. Initially binary data input b_k is converted into **bipolar NRZ format**. This NRZ signal is then given to the mixer. As this is the input data sequence, it is narrowband signal.

2. The other input to the mixer is the wideband PN code c(t). Modulo-2 addition (**EX-ORing operation**) is carried out and output is provided to the next stage which is **PSK modulator**. This output of the multiplier is known as 'spreading code'. It is now the wideband signal whose bandwidth is almost equal to the PN code signal.

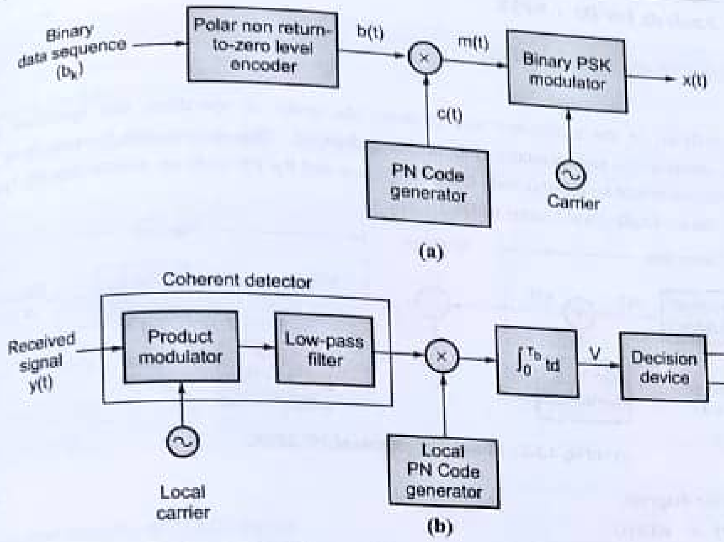
3. Therefore, the transmitted signal is DS/BPSK signal. In BPSK, two phases, **0** and π are present depending on the polarity of the data sequence. The waveforms are as shown in Fig. 4.5.1(c).

4. ***Refer Fig. 4.5.1 (b).*** At the receiver, the received signal is first given to the product modulator and low pass filter combination. The **cut-off frequency** of the low pass filter is equal to the original message signal. Coherent detection is carried out for detection of the spreading code from the BPSK output. The output of the coherent detector is the spreading code.

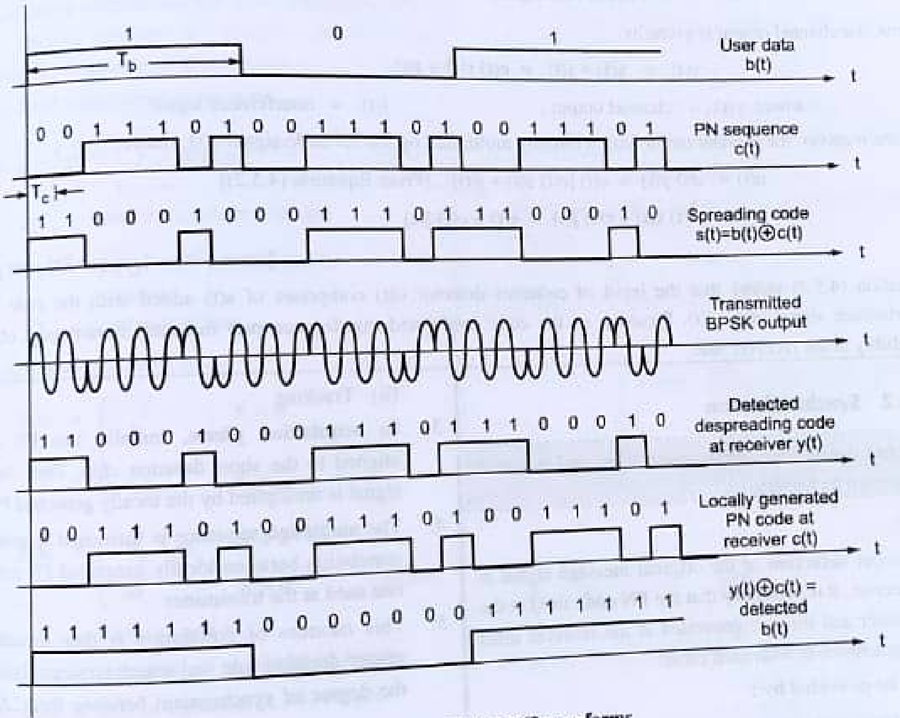
5. This spreading code and locally generated PN code are then multiplied. Digitally, modulo-2 addition or EX-OR operation is carried out. The condition of this PN code generation is that, it has to be same as that of the PN code used at the transmitter. This process is called as **'de-spreading'**.

6. After the de-spreading of the signal, it is given to the integrator. Integration is carried out for one bit duration.

7. Integrator's output is provided to the decision device. The threshold is set for **logic '1'** and **logic '0'**. If the incoming bit is above the threshold, it is declared as **logic '1'** otherwise **logic '0'**. If the bit is exactly equal to the threshold, then the decision device takes the decision randomly.



(1F10)Fig. 4.5.1 : (a) DSSS transmitter and (b) DSSS receiver



(1F11)Fig. 4.5.1(c) : DS / BPSK waveforms

Unit IV End Sem.

Sr. No.	Parameter	BASK	BPSK	DPSK	QPSK	M-ary PSK	QAM	BFSK	M-ary FSK
5.	Bandwidth	$2f_b$	$2f_b$	f_b	f_b	$\frac{2f_b}{N}$	$\frac{2f_b}{N}$	$4f_b$	$\frac{2^{N+1}}{N} f_b$
6.	Error probability	$\frac{1}{2}\text{erfc}\sqrt{\frac{E_b}{4N_0}}$	$\frac{1}{2}\text{erfc}\sqrt{\frac{E_b}{N_0}}$	$\frac{1}{2}e^{\frac{-E_b}{N_0}}$	$\text{erfc}\sqrt{\frac{E_b}{N_0}}$	$\text{erfc}\left[\sqrt{\frac{E_b}{N_0}}\sin\frac{\pi}{M}\right]$	$\text{erfc}\sqrt{\frac{E_b}{N_0}}$	$\frac{1}{2}\text{erfc}\sqrt{\frac{0.6E}{N_0}}$	$\frac{M-1}{2}\text{erfc}\sqrt{\frac{E_s}{2N_0}}$
7.	Euclidean distanced	$\sqrt{E_b}$	$2\sqrt{E_b}$		$2\sqrt{E_b}$	$2\sqrt{E_b}\sin\frac{\pi}{M}$	$\sqrt{0.4E_b}$ for M = 16	$\sqrt{2E_b}$	$\sqrt{2NE_b}$
8.	Symbol duration (T_s)	T_b	T_b	$2T_b$	$2T_b$	NT_b	NT_b	T_b	NT_b
9.	Applications	low data rate applications	Telemetry		telephony, satellite communication	Satellite communication	Microwave digital radio, link, DVB	Paging services, cordless telephony	Paging services, cordless telephony

3.6 INTER SYMBOL INTERFERENCE (ISI)

3.6.1 Concept of ISI

UQ. What is Inter Symbol Interference (ISI)?

SPPU - (E&TC Sem. 5) May 15(In Sem.), May 16(In Sem.), Dec. 16, 2 Marks

Ans. :

A communication channel is always band limited. Therefore, it always spreads or disperses the waveform passing through it.

When the channel bandwidth is greater than pulse bandwidth, spreading of the pulse is small, whereas when the channel bandwidth is close to pulse bandwidth, the spreading of the pulse exceeds the symbol duration.

Due to this excessive spreading of pulse, it overlaps the adjacent symbol pulses. Such overlapping of pulses is known as ISI.

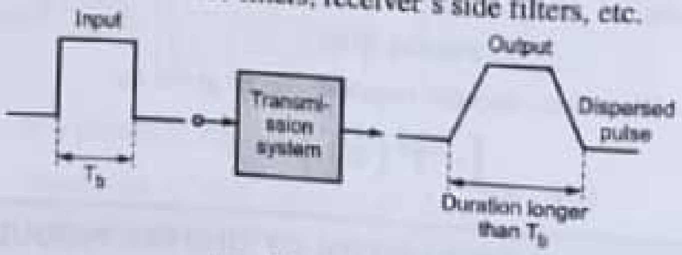
Usually to combat noise, signal power is increased but this does not work in case of ISI. Sometimes effect of ISI is really severe and degrades the signal too much.

Definition : Spreading of a pulse beyond its interval (T_b sec.) will cause it to interfere with neighbouring pulses. This is known as intersymbol interference.

Even in the absence of noise, the effects of filtering and

3.6.1(A) Illustration of Concept of ISI

- Refer Fig. 3.6.1. It shows typical baseband digital system. There are many filters present in the system like transmitter's side filters, receiver's side filters, etc.



(1844)Fig. 3.6.1 : Concept of ISI

- At the transmitter, signal is band limited using anti aliasing filter. The channels have distributed r
- They can distort the pulse. Wireless system channels. They may have unwanted fi distorting the pulses.
- Receiver is always refe qualizin needs to compensate ise and effects.
- Hence the combined tr tion for the system can be writte

$$H(f) = \quad (f)\ H_r($$

where $H_t(f)$ = Transmi ansfer
$H_r(f)$ = Receive r func

- Due to the effects of ing, t pulses can overlap one grad

Soln. :

Given :

(1) $x_{01}(T) = +V$

(2) $x_{02}(T) = 0$

$$\text{Decision Threshold} = \frac{x_{01}(T) + x_{02}(T)}{2} = \frac{V+0}{2} = \frac{V}{2}$$

2.5 BINARY PHASE SHIFT KEYING (BPSK)

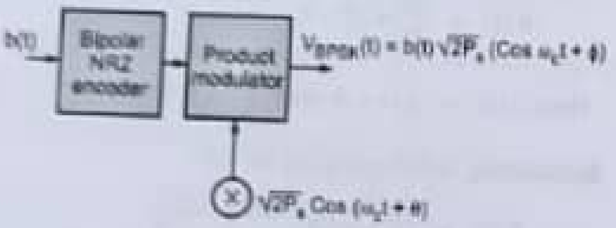
Generation and detection of BPSK

2.5.1 BPSK Transmitter

GQ. Explain BPSK transmitter with neat block diagram.

Ans. :

Refer Fig. 2.5.1. It shows block diagram of BPSK generator and coherent BPSK receiver.



(1E20)Fig. 2.5.1 : Block diagram of BPSK generator

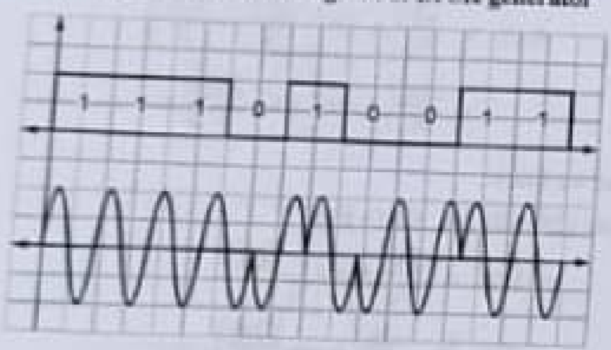


Fig. 2.5.1(a) : Waveforms of BPSK

(1) Polar NRZ encoder
(2) Product modulator

(1) Polar NRZ encoder

- It converts the input binary data sequence into polar NRZ format.
- In this, binary symbols '1' and '0' are represented by $\pm\sqrt{2P_s}$. P_s is the signal power of sinusoidal signal with amplitude 'A'.

$$\therefore P_s = \frac{1}{2}\cdot A^2$$

$$\therefore A = \sqrt{2P_s}$$

$$\therefore \textbf{Logic 1} \rightarrow +\sqrt{2P_s}$$

$$\textbf{Logic 0} \rightarrow -\sqrt{2P_s}$$

(2) Product modulator

- The polar NRZ symbols are multiplied with sinusoidal carrier. The desired BPSK output has fixed phase when the data is at one level.
- It changes the phase by 180° for other level change. Thus, the mathematical expression for the output of BPSK signal can be modelled.

Mathematical expression

$$V_{BPSK}(t) = \sqrt{2P_s}\cdot\cos(\omega_c t) \quad 0^\circ \text{ phase}$$

$$\text{or } V_{BPSK}(t) = \sqrt{2P_s}\cdot\cos(\omega_c t + \pi) \quad \pi \text{ phase}$$

$$= -\sqrt{2P_s}\cdot\cos(\omega_c t)$$

Hence in a general form,

$$V_{BPSK}(t) = b(t)\sqrt{2P_s}\cdot\cos\omega_c t$$

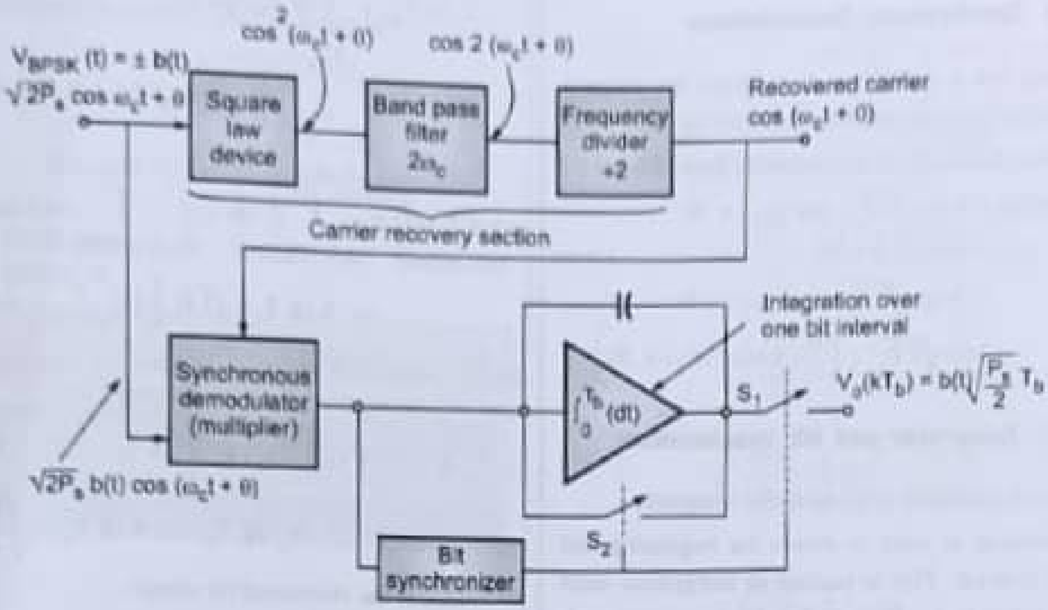
It is the expression for output BPSK.

• NOTES •

2.5.2 BPSK Receiver

GQ. Explain coherent BPSK receiver with neat block diagram.

Ans. :



(1E24)Fig. 2.5.2 : Block diagram of BPSK receiver

- *Refer Fig. 2.5.2.* It shows block diagram of BPSK generator and coherent BPSK receiver. The received signal at BPSK receiver is,

$$V_{BPSK}(t) = b(t)\sqrt{2P_s}\cdot\cos(\omega_c t + \theta)$$

$$= b(t)\sqrt{2P_s}\cdot\cos\left[\omega_c\left(t + \frac{\theta}{\omega_c}\right)\right]$$

θ is a fixed phase shift. It refers to time delay $\left(\frac{\theta}{\omega_c}\right)$.

- The time delay $\left(\frac{\theta}{\omega_c}\right)$ depends on majorly two factors :
  (1) Distance between transmitter and receiver.
  (2) Phase shift produced by the amplifiers in the front ends of the receivers.
- In coherent detection techniques, practically synchronous demodulator is needed. It requires separate carrier recovery along with original received signal.

2.5.2(A) Carrier Recovery Section

1. Square law device
2. Band pass filter ($2f_c$)
3. Frequency divider (÷ 2)

1. Square law device

- It is provided with the received BPSK output. Hence, input of square law device = $V_{BPSK}(t)$.
- It will square the received signal. Hence output of square law device

$= \cos^2(\omega_c t + \theta)$ ...[Amplitude is ignored to avoid complications in simplifications]

$$\therefore \cos^2(\omega_c t + \theta) = \underbrace{\frac{1}{2}}_{\text{DC component}} + \underbrace{\frac{1}{2}\cos 2(\omega_c t + \theta)}_{\text{Second harmonic of carrier}}$$

2. Band pass filter

- The band pass filter used has the

Cut off frequency = $2f_c$

- The filter removes the DC component. Hence output of band pass filter = $[\cos^2(\omega_c t + \theta)]$

3. Frequency divider

- It consists of flip-flop and narrowband filter tuned to f_c. It divides the incoming signal frequency by a factor of 2.

- Hence, Output of frequency divider $= \cos(\omega_c t + \theta)$

$\downarrow$

Carrier is recovered.

2.5.2(B) Synchronous Demodulator

- It is nothing but a multiplier. It is given the original BPSK (t) signal and the recovered carrier signal.
- Mathematical equations at synchronous demodulator :

$$\text{Output of multiplier} = b(t)\sqrt{2P_s}\cdot\cos(\omega_c t + \theta)\,\cos(\omega_c t + \theta)$$

$$= b(t)\sqrt{2P_s}\cdot\cos^2(\omega_c t + \theta)$$

$$= b(t)\sqrt{2P_s}\cdot\left[\frac{1}{2} + \frac{1}{2}\cos^2(\omega_c t + \theta)\right]$$

2.5.2(C) Integrator and Bit Synchronizer

- The output of multiplier is given to the integrator.
- Bit synchronizer is used to detect the beginning and end of bit interval. This is needed as integration must be carried out over one bit duration. It is integrator-and-dump switch combination.

Mathematical equations at output of integrator

Let $V_o(kT_b)$ be the output voltage of integrator after each bit interval. It has duration from $(k-1)T_b$ to kT_b.

$$\therefore\ V_o(kT_b) = b(kT_b)\sqrt{2P_s}\cdot\int_{(k-1)T_b}^{kT_b}\left[\frac{1}{2} + \frac{1}{2}\cos 2(\omega_c t + \theta)\right]dt$$

$$= b(kT_b)\sqrt{2P_s}\int_{(k-1)T_b}^{kT_b}\frac{1}{2}\cdot dt + \frac{1}{2}\int_{(k-1)T_b}^{kT_b}\cos 2(\omega_c t + \theta)\,dt$$

$$= b(kT_b)\sqrt{2P_s}\,\frac{1}{2}[t]_{(k-1)T_b}^{kT_b} + 0$$

Second term $= 0$ because, integral of sinusoidal function over whole number of cycles is always zero.

$$\therefore\ V_o(kT_b) = b(kT_b)\sqrt{2P_s}\cdot\frac{1}{2}\,[kT_b - kT_b + T_b]$$

$$\therefore\ V_o(kT_b) = b(kT_b)\sqrt{\frac{P_s}{2}}\cdot T_b$$

This is the recovered bit stream.

2.5.3 Power Spectrum of Binary PSK System

GQ. Describe power spectrum of BPSK system.

Ans. :

1. The BPSK waveform has b (t) in NRZ binary format. Hence, the binary NRZ waveform has two levels $+\sqrt{P_s}$ and $-\sqrt{P_s}$. Hence, Fourier transform of this waveform is

$$S_b(f) = P_s T_b\left(\frac{\sin \pi f T_b}{\pi f T_b}\right)^2$$

2. The BPSK waveform has NRZ waveform multiplied with carrier $\sqrt{2}\cos\omega_c t$. Hence the power spectral density of BPSK signal is given as,

$$S_{BPSK}(f) = \frac{P_s T_b}{2}\left\{\left[\frac{\sin \pi (f - f_c) T_b}{\pi (f - f_c) T_b}\right]^2 + \left[\frac{\sin \pi (f + f_c) T_b}{\pi (f + f_c) T_b}\right]^2\right\}$$

It is plotted in Fig. 2.5.3.

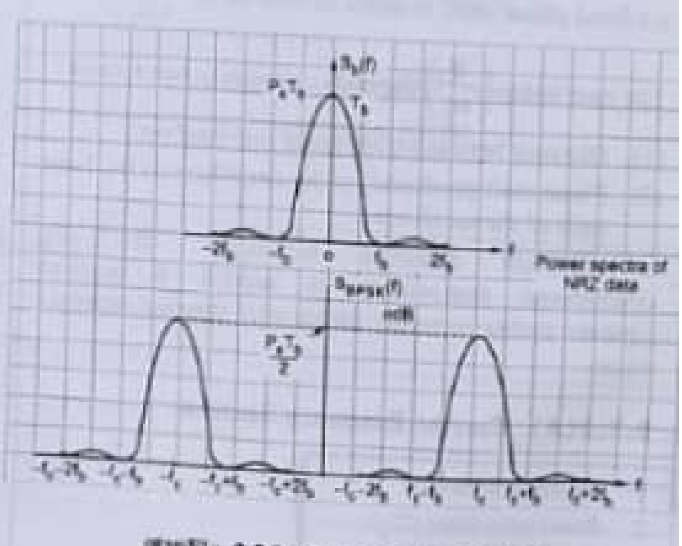


Fig. 2.5.3 : Power spectra of BPSK

2.5.4 Bandwidth of BPSK

GQ. State bandwidth of BPSK system.

Ans. :

Bandwidth = Higher frequency – Lower frequency

$$\therefore \quad B = (f_c + f_b) - (f_c - f_b)$$

$$\therefore \quad B = 2 f_b$$

$$\text{Where} \quad f_b = \frac{1}{T_b} = \frac{1}{\text{bit interval}}$$

2.5.5 Probability of Error for Coherent BPSK

UQ. Derive an expression of error probability of BPSK using matched filter.

SPPU - (E&TC-Sem. 5) May 15, May 16, 8 Marks

UQ. Derive the expression for error probability of BPSK receiver. **SPPU - (E&TC-Sem. 5) May 17, 8 Marks**

Ans. :

In BPSK:

1. For binary 1, $V_{BPSK}(t) = x_1(t) = \sqrt{2P_s} \cdot \cos \omega_c t$
2. For binary 0, $V_{BPSK}(t) = x_2(t) = -\sqrt{2P_s} \cdot \cos \omega_c t$

Hence, $x_2(t) = -x_1(t)$

- As we know, the probability of error for matched filter is, $P_e = \frac{1}{2} \operatorname{erfc}\left[\frac{x_{01}(T) - x_{02}(T)}{2 \cdot \sqrt{2} \cdot \sigma}\right]$ ...(2.5.1)

$$\text{Where} \left[\frac{x_{01}(T) - x_{02}(T)}{\sigma}\right]_{max} = \frac{2}{N_0}\int_0^T x^2(t)\, dt \quad \text{...(2.5.2)}$$

Here $x(t) = x_1(t) - x_2(t)$

For BPSK, $x_2(t) = -x_1(t)$

$$\therefore \quad x(t) = x_1(t) - [-x_1(t)] = 2 x_1(t) \quad \text{...(2.5.3)}$$

Substituting Equation (2.5.3) in Equation (2.5.2) we get, $= \frac{2}{N_0}\int_0^T x^2(t)\, dt$

$$= \frac{2}{N_0}\int_0^T [2 x_1(t)]^2\, dt = \frac{2}{N_0}\int_0^T 4 x_1^2(t)\, dt$$

$$= \frac{8}{N_0}\int_0^T x_1^2(t)\, dt \quad \text{...(2.5.4)}$$

But $x_1(t) = \sqrt{2P_s} \cdot \cos \omega_c t$, and $T = T_b$ = bit interval

$$\therefore \quad = \frac{8}{N_0}\int_0^{T_b} 2 P_s \cos^2 \omega_c t\, dt \quad \text{...(2.5.5)}$$

$$= \frac{16 P_s}{N_0}\int_0^{T_b} \cos^2 \omega_c t \cdot dt$$

$$\text{Now,} \quad \cos^2 \omega_c t = \frac{1 + \cos 2\omega_c t}{2}$$

$$\text{Hence} \quad = \frac{16 P_s}{N_0}\int_0^{T_b}\left[\frac{1 + \cos 2\omega_c t}{2}\right] dt$$

$$= \frac{16 P_s}{N_0}\left[\int_0^{T_b} \frac{1}{2}\, dt + \frac{1}{2}\int_0^{T_b} \cos 2\omega_c t\, dt\right]$$

$$= \frac{16 P_s}{N_0} \times \frac{1}{2} [t]_0^{T_b} + \frac{16 P_s}{N_0}$$

...(2.5.6)

$$= \left\{\frac{8 P_s}{N_0} \times T_b\right\} + 0 \quad \text{...(2.5.7)}$$

Sinusoidal function of whole numbers is zero.

$$\therefore \quad = \frac{8 P_s \cdot T_b}{N_0} \quad \text{...(2.5.8)}$$

But $P_s \cdot T_b = E_b$ = energy in bit duration.

$$\therefore \left[\frac{x_{01}(T) - x_{02}(T)}{\sigma}\right]^2 = \frac{8 E_b}{N_0} \quad \text{...(2.5.9)}$$

- Substituting Equation (2.5.9) in Equation (2.5.1) we get,

$$P_e = \frac{1}{2} \operatorname{erfc}\left[\frac{1}{2\sqrt{2}}\sqrt{\frac{8 E_b}{N_0}}\right]$$

$$= \frac{1}{2} \operatorname{erfc}\left[\frac{2\sqrt{2}}{2\sqrt{2}}\sqrt{\frac{E_b}{N_0}}\right]$$

$$P_{e(BPSK)} = \frac{1}{2} \operatorname{erfc}\left[\sqrt{\frac{E_b}{N_0}}\right]$$

...Error probability of BPSK ...(2.5.10)

As we know, $\operatorname{erfc}(u) = 2\,Q(\sqrt{2} \cdot u)$

$$\therefore \quad \text{Let} \quad u = \sqrt{\frac{E_b}{N_0}}$$

$$\operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right) = 2\,Q\left\{\sqrt{2} \cdot \sqrt{\frac{E_b}{N_0}}\right\}$$

2.6 DIFFERENTIAL PSK

(For reading purpose only..)

Drawback of BPSK receiver

- In BPSK receiver shown in Fig. 2.5.2 square law device is used.
- It squares the received signal. If the received signal is $\left[-b(t)\sqrt{2P_s}\cos\omega_c t\right]$, the recovered carrier would still be the same as it is for $\left[+b(t)\sqrt{2P_s}\cos\omega_c t\right]$.
- As square device will make no difference with (– b(t)) term, it is difficult to determine + b(t) and – b(t).
- Solutions for drawback of BPSK :

  (i) DPSK (ii) DEPSK

Unit II In Sem.

2.6.1 Generation and Detection of DPSK

UQ. Draw the block diagram of DPSK transmitter and explain its operation with proper waveforms.
SPPU - (E&TC-Sem. 5) Dec. 14, 6 Marks

UQ. Explain with the help of block diagram and waveforms DPSK modulation. **SPPU - (E&TC-Sem. 5) May 16, 8 Marks**

Ans. :

DPSK Transmitter

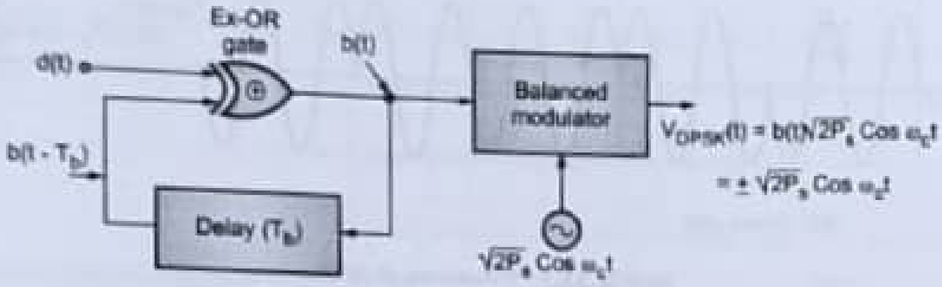


Fig. 2.6.1: Block diagram of DPSK transmitter

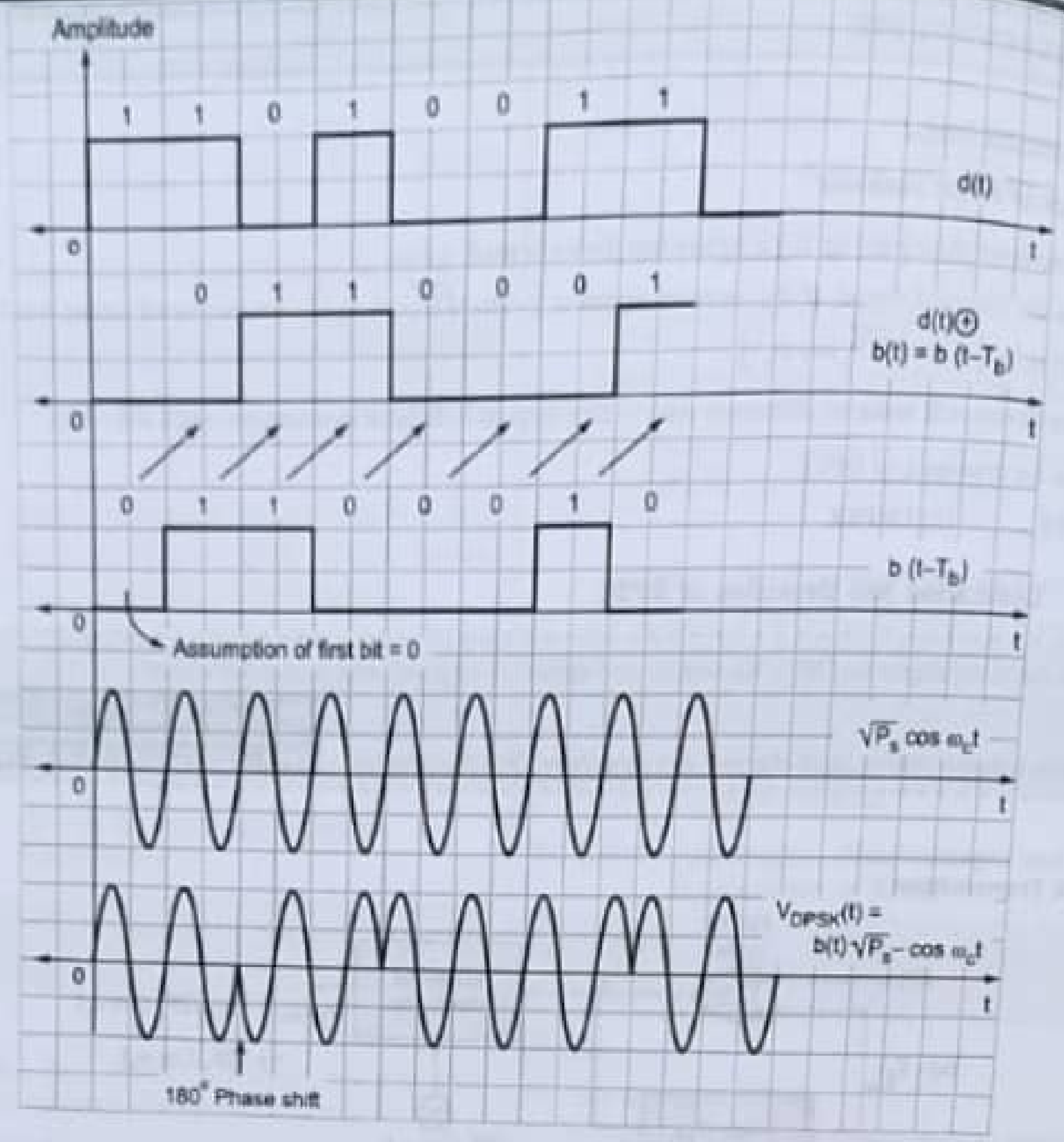
- It comprises of Ex-OR gate, delay block and balanced modulator. The data stream d (t) is applied as one of the input of Ex-OR Gate. The other input of Ex-OR gate is provided with delayed version of output of Ex-OR gate, i.e., **b (t – T_b)**. One bit delay is added to the output of Ex-OR gate. Truth table of Ex-OR gate is as shown in Table 2.6.1.

Table 2.6.1 : Truth table of Ex-OR gate

INPUT		OUTPUT
A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

$Y = A \oplus B$

- The logic waveforms are shown in Fig. 2.6.2. It shows that for d (t) = 0, phase does not change and for d (t) = 1, phase changes by 180°. Initially for the 1[st] interval b (t) = 0 (assumed). This b (t) obtained is then applied to the balanced modulator. The carrier $\left[\sqrt{2P_s}\cos\omega_c t\right]$ is applied to the balanced modulator. At the output of balanced modulator, DPSK output is obtained.



(186)Fig. 2.6.2 : Waveforms of DPSK

2.6.2 Mathematical Expression

GQ. State the mathematical expression for DPSK.

Ans. :

$$V_{DPSK}(t) = b(t)\sqrt{2P_s}\cdot\cos\omega_c t$$

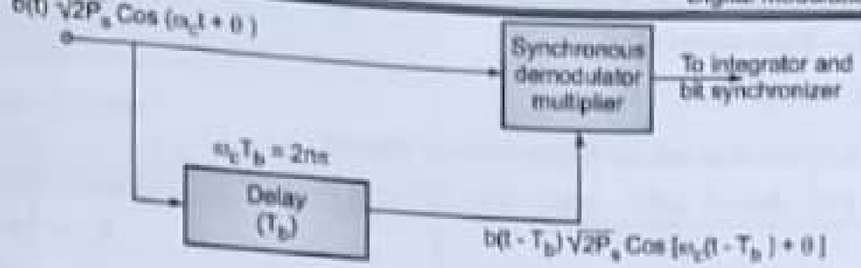
$$= \pm\sqrt{2P_s}\cdot\cos\omega_c t \qquad \dots (2.6.1)$$

DPSK receiver

GQ. Explain detection of DPSK system with neat waveforms and mathematical analysis.

Ans. :

- *Refer Fig. 2.6.3.* It shows block diagram of DPSK receiver. It comprises of (1) delay block, (2) the synchronous multiplier and (e) integrator and bit synchronizer combination.



(1E28)Fig. 2.6.3 : Block diagram of DPSK receiver

- The received signal is, $V_{DPSK}(t) = b(t)\sqrt{2P_s}\cdot\cos(\omega_c t + \theta)$ ...(2.6.2)

  This received signal is applied to the synchronous multiplier and the delay block. Delay block adds the delay of one bit duration.

  Hence, Output of delay block $= b(t - T_b)\sqrt{2P_s}\cdot\cos[\omega_c(t - T_b) + \theta]$ ...(2.6.3)

- At the multiplier

$$= \underbrace{b(t)\sqrt{2P_s}\cdot\cos(\omega_c t + \theta)}_{\text{Received DPSK signal}} \times \underbrace{b(t-t_b)\sqrt{2P_s}\cdot\cos[\omega_c(t-T_b) + \theta]}_{\text{delayed DPSK signal}} \quad ...(2.6.4)$$

$$= \mathbf{b(t)\, b(t - T_b)\,(2P_s)\cos(\omega_c t + \theta)\cos[\omega_c(t - T_b) + \theta]} \quad ...(2.6.5)$$

$\cos A\cdot\cos B = \frac{1}{2}[\cos(A + B) + \cos(A - B)]$ can be used.

Hence, $\cos(\omega_c t + \theta)\cdot\cos[\omega_c(t - T_b) + \theta] = \cos(\omega_c t + \theta)\cdot\cos(\omega_c t - \omega_c T_b + \theta)$

$$= \frac{1}{2}\{\cos[\omega_c t + \theta + \omega_c t - \omega_c t_b + \theta] + \cos[\omega_c t + \theta - \omega_c t + \omega_c T_b - \theta]\}$$

$$= \frac{1}{2}\{\cos[2\omega_c t - \omega_c T_b + 2\theta] + \cos(\omega_c T_b)\}$$

$$= \frac{1}{2}\left\{\cos\left[2\omega_c\left(t - \frac{T_b}{2}\right) + 2\theta\right] + \cos(\omega_c T_b)\right\} \quad ....(2.6.6)$$

Substituting Equation (2.6.6) in Equation (2.6.5) we get,

$$\text{The multiplier output} = b(t)\cdot b(t - T_b)(2P_s)\frac{1}{2}\left\{\cos\left[2\omega_c\left(t - \frac{T_b}{2}\right)\right] + 2\theta + \cos(\omega_c T_b)\right\} \quad ...(2.6.7)$$

$$= b(t)\, b(t - T_b)\cdot P_s\left\{\cos\left[2\omega_c\left(t - \frac{T_b}{2}\right) + 2\theta\right] + \cos(\omega_c T_b)\right\}$$

- It is applied to integrator and bit synchroniser combination. Let the output of integrator be $V_0(kT_b)$.

$$\therefore V_0(kT_b) = \int_{(k-1)T_b}^{kT_b} b(t)\cdot b(t - T_b)P_s\left\{\cos\left[2\omega_c\left(t - \frac{T_b}{2}\right) + 2\theta\right]\right\}dt + \int_{(k-1)T_b}^{kT_b} b(t)\cdot b(t - T_b)P_s\cos(\omega_c T_b)\,dt \quad ...(2.6.8)$$

Put $\omega_c T_b = 2n\pi$ $\quad\therefore \cos 2n\pi = 1$

- First term of integration = 0, as integration of sinusoidal function over whole numbers = 0.

$$\therefore V_0(kT_b) = \int_{(k-1)T_b}^{kT_b} b(t)\cdot b(t - T_b)P_s\cdot 1\,dt = b(t)\cdot b(t - T_b)P_s\cdot[t]_{(k-1)T_b}^{kT_b} \quad ...(2.6.9)$$

Let $t = k^{th}$ interval $\quad\therefore t = kT_b$

$$\therefore V_0(kT_b) = b(kT_b)\, b(kT_b - 1)\cdot P_s[kT_b - (k - 1)T_b] \quad ...(2.6.10)$$

$$= b(kT_b)\cdot b(kT_b - 1)\cdot P_s T_b$$

But $P_s \cdot T_b = E_b$ = Signal energy of bit

$$\therefore V_0(kT_b) = E_b \cdot b(kT_b) \cdot b(kT_b - 1) \quad ...(2.6.11)$$

- From Equation (2.6.11) it is clear that, bit d(t) can be easily detected.

If $b(t) \cdot b(t - T_b) = 1$ V, $d(t) = 0$ and if $b(t) \cdot b(t - T_b) = -1$ V, $d(t) = 1$

2.6.3 Bandwidth of DPSK

GQ. State bandwidth of DPSK.

Ans. :

In DPSK, previous bit is used to decide the phase shift of next bit. Hence the symbol is considered to have two bits. Hence the duration of symbol can be,

$$\text{Symbol duration} = T = 2T_b$$

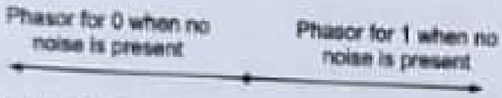
$$\therefore \text{Bandwidth} = \frac{2}{T} = \frac{2}{2T_b} = \frac{1}{T_b}$$

$$\therefore B = f_b$$

2.6.4 Error Probability of DPSK

GQ. State error probability of DPSK.

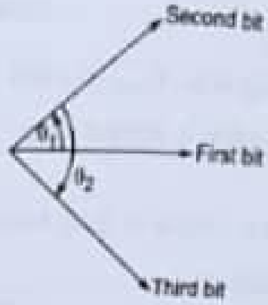
Ans. :



(1E22)Fig. 2.6.4(a) : **Phasor diagram of DPSK when no noise is present**

- Refer the phasor diagrams of DPSK plotted in Fig. 2.6.4(a). When noise is absent, phase = 0 or π
- Hence decision boundary = $\frac{0+\pi}{2} = \frac{\pi}{2}$. Hence '0' is detected if phase $> \frac{\pi}{2}$ and '1' is detected if phase $< \frac{\pi}{2}$
- *Refer Fig. 2.6.4(b)* - Noise is present. DPSK receiver compares bit (2) with bit (1) and as angle $\theta_1 < \pi/2$, bit (2) = 1.
- Similarly, when bit (3) is compared with bit (2), angle $\theta > \frac{\pi}{2}$. Hence, bit (3) = 0. Hence, DPSK receiver always takes previous bit as a reference. The error probability is given as,

$$P_e = \frac{1}{2} e - (E_b / N_0)$$



(1E23)Fig. 2.6.4(b) : **Phasor diagram of DPSK when noise is present**

2.6.5 Advantages, Disadvantages and Applications of DBPSK

GQ. State advantages, disadvantages and applications.

UQ. What advantage does DBPSK have over conventional PSK ? (MU - Q. 1(2), May 18, 2 Marks)

Ans. :

Advantages

(1) It does not need the locally recovered carrier signal at the receiver. Hence compared to BPSK detection, circuit of DPSK detection is easy.

(2) Bandwidth of DPSK is lesser than BPSK.

Disadvantages

(1) The probability of error is higher than BPSK.

(2) Receiver makes use of two successive bits for detection of single bit. Hence, error in one bit can be carried forward to next bits. Hence, error rate increases.

(3) Noise immunity is lower as compared to BPSK.

Application : Bluetooth technology.

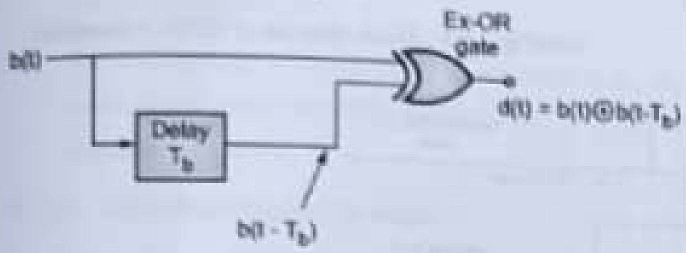
2.7 DIFFERENTIALLY ENCODED PSK (DEPSK)

2.7.1 Drawback of DPSK

GQ. Explain DEPSK system in detail.

Ans. :

- The DPSK demodulator needs a device which operates at carrier frequency. Also it is able to provide delay T_b.
- It avoids the need of extra hardware requirement in DPSK. *Refer Fig. 2.7.1*. It shows block diagram of detection of DEPSK signal.



(1E20)Fig. 2.7.1 : Block diagram of differentially encoded PSK (DEPSK)

- The transmitter of DPSK and DEPSK are identical, but the detection method is different. As shown in Fig. 2.7.1, the detection method resembles BPSK receiver except the last block. It consists of Ex-OR gate and delay block. The recovered signal b(t) is applied at the input of Ex-OR gate and to the delay block.
- Hence, two inputs of Ex-OR gate are b(t) and $b(t-T_b)$. Therefore, output of Ex-OR gate d(t).

  i.e. $\mathbf{d(t) = b(t) \oplus b(t-T_b)}$.

  Hence, $d(t) = 0$ if $b(t) = b(t-T_b)$

  And, $d(t) = 1$ if $b(t) = \overline{b(t-T_b)}$

2.7.2 Errors in DEPSK System

GQ. Explain the errors in DEPSK system with example.

Ans. :

In DEPSK, errors always occur in pairs. This is, because in DEPSK a definite hard decision is given for bit interval b (t).

Example :

```
               b(k) : 0 1 1 0 1 1 0 0
             b(k-1) :   0 1 1 0 1 1 0 0    One error
d(t) = b(k) ⊕ b(k-1) :   1 0 1 1 0 1 0
              b'(k) : 0 1 1   1 1 0 0
            b'(k-1) :   0 1 1 1 1 1 0 0
d'(k) = b'(k) ⊕ b'(k-1) : 1 0 0 0 0 1 0
                            Two errors
```

(1E4)Fig. 2.7.2 : Errors in DEPSK

2.7.3 Advantages and Disadvantages of DEPSK

GQ. State advantages and disadvantages of DEPSK.

Ans. :

Advantages

1. As coherent detection is used, probability of error reduces. Hence, probability of error is lesser than DPSK system.
2. AT DEPSK receiver, delay block operates at baseband frequency and not on carrier frequency, like in DPSK. It reduces the size and cost of hardware.

Disadvantages

1. Receiver is complex. 2. Errors always occur in pairs.

2.8 QUADRATURE PHASE SHIFT KEYING (QPSK)

Working Principle

GQ. State the working principle of QPSK.

Ans. :

(i) In QPSK or 4-PSK system, two bits are grouped together to form a symbol. Hence at the output, 4 different combinations of binary bits [**00, 01, 10** and **11**] are available.

(ii) These four symbols are represented with four different phases. Hence, changes in phases of carrier carry the information in QPSK.

Unit II In Sem.

Ex. 2.8.1 : Sketch PSK and QPSK signals for the input bit sequence 10011010.

☑ **Soln. :**

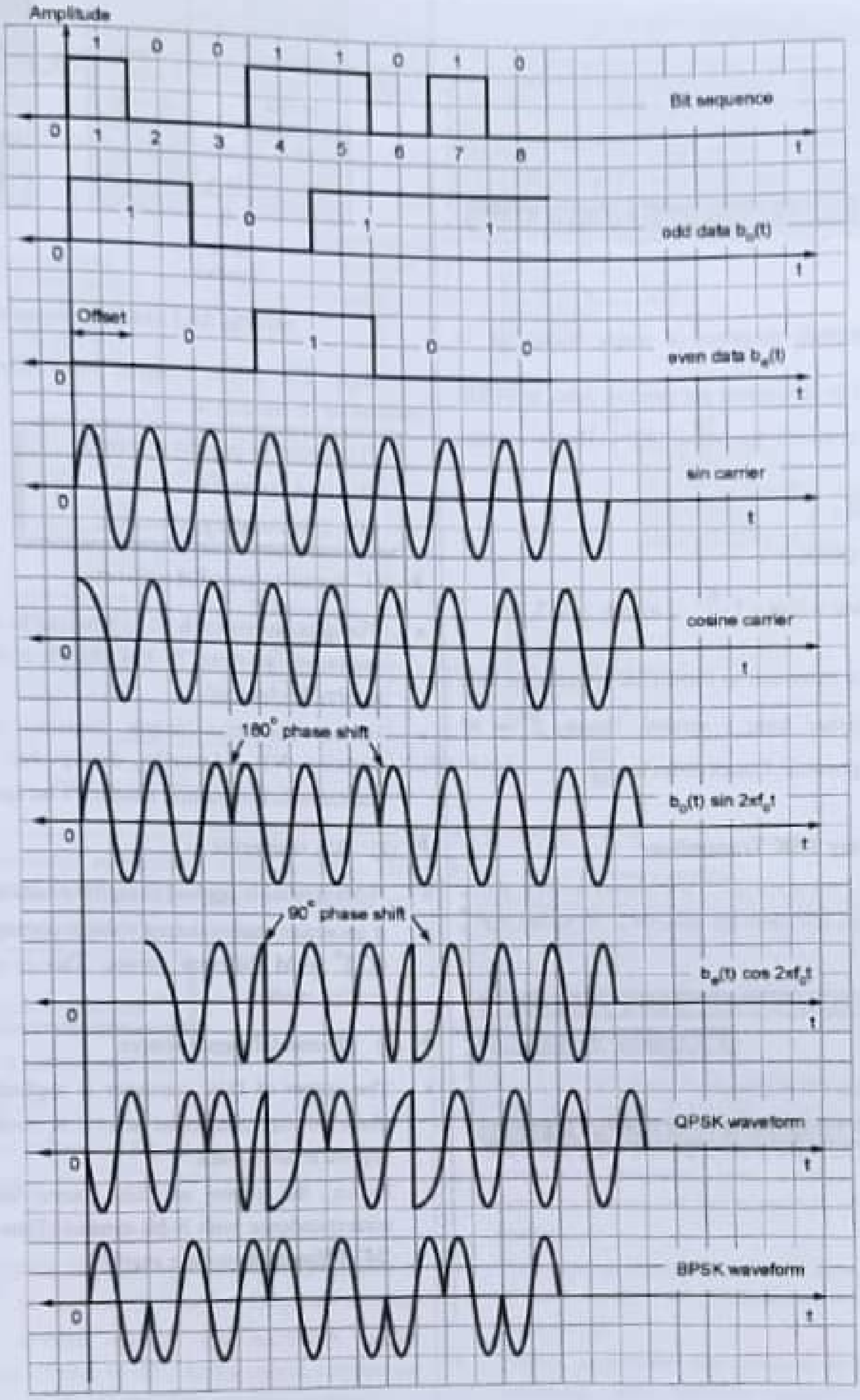


Fig. P. 2.8.1

(1) D Flip-Flop (Delay)	(2) T Flip-Flop (Toggle)
(3) Multiplier	(4) Adder.

(1) D Flip-Flop (Delay Flip-Flop)

- Type D flip-flop has only single input terminal. It delays the input by one clock period. It gives output at active edge of the clock signal.
- In case of QPSK transmitter, type D flip-flop is used to separate the data bit stream d (t) in odd data bits $b_o(t)$ and even data bits $b_e(t)$.
- The trailing edge of clock cycle is assumed to be the active edge. Hence at every trailing edge D–input is shifted to 'Q'–output.

(2) T flip-flop (Toggle flip-flop)

- It is given clock signal whose period is T_b. The toggle flip-flop generates odd clock and even clock. They are given to respective D flip flops.
- The generated odd clocks and even clocks have **bit duration = $2T_b$**. As the bit stream b (t) is applied to D flip-flop and also it is given clock of $2T_b$ period, the output waveform obtained has **bit period = $2T_b$**.
- Hence, data is separated in odd and even data streams. Resulting waveforms are shown in Fig. 2.8.2(b) and Fig. 2.8.2 (c).

(3) Multiplier

- Two different multiplier devices are used in QPSK transmitter. They are provided with $b_o(t)$, $b_e(t)$ and sinusoidal carriers, respectively.

  Hence at the output of multiplier 1,
  $S_e(t) = [b_e(t)\sqrt{P_s}\cdot\cos\omega_c t]$ is obtained. Similarly at the output of multiplier 2, $S_o(t) = [b_o(t)\sqrt{P_s}\cdot\sin\omega_c t]$ signal is obtained.

(4) Adder

$S_e(t)$ and $S_o(t)$ signals are added in the adder block. At the output, QPSK signal is obtained.

2.8.1(A) Mathematical Expression for QPSK

$$V_{QPSK}(t) = \sqrt{P_s}\cdot b_o(t)\sin\omega_c t + \sqrt{P_s}\cdot b_e(t)\cos\omega_c t$$
$$= S_o(t) + S_e(t)$$

In a generalized form,

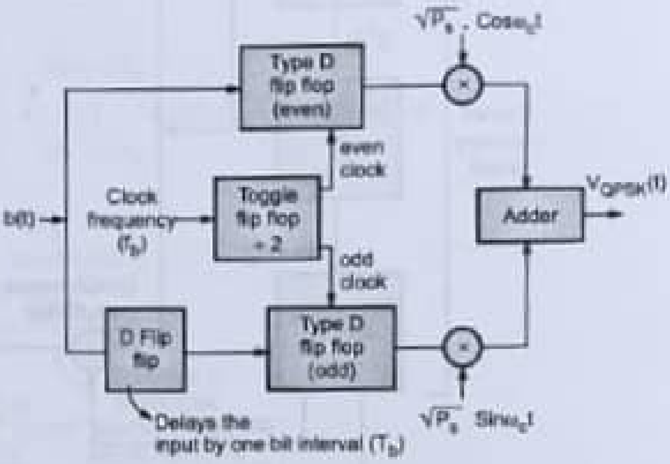
$$V_{QPSK}(t) = \sqrt{2P_s}\cdot\cos\left[\omega_c t + (2m+1)\frac{\pi}{4}\right]$$
$$m = 0, 1, 2, 3.$$

- As shown in waveforms Fig. 2.8.2, offset is occurred in the output. Hence, this system is known as 'offset QPSK'.
- To avoid this offset, few changes in already existing OQPSK transmitter are done. The newly developed system is known as 'Non-offset QPSK'.

2.8.2 Non-offset QPSK

GQ. Explain non-offset QPSK with block diagram.

Ans. :



(1B9)Fig. 2.8.3 : Block diagram of non-offset QPSK transmitter

- *Refer Fig. 2.8.3.* It shows Non-offset QPSK Transmitter. Only one additional D flip-flop is added before odd flip-flop. It can be added before even flip-flop also, instead of odd one.
- This added D flip-flop is driven by the clock of duration T_b. Hence, odd bit stream will be delayed by one bit period. As a result, both the bit streams will occur serially. Principle of working remains same.

Unit II In Sem.

2.8.2(A) Problems associated with Non-Offset QPSK

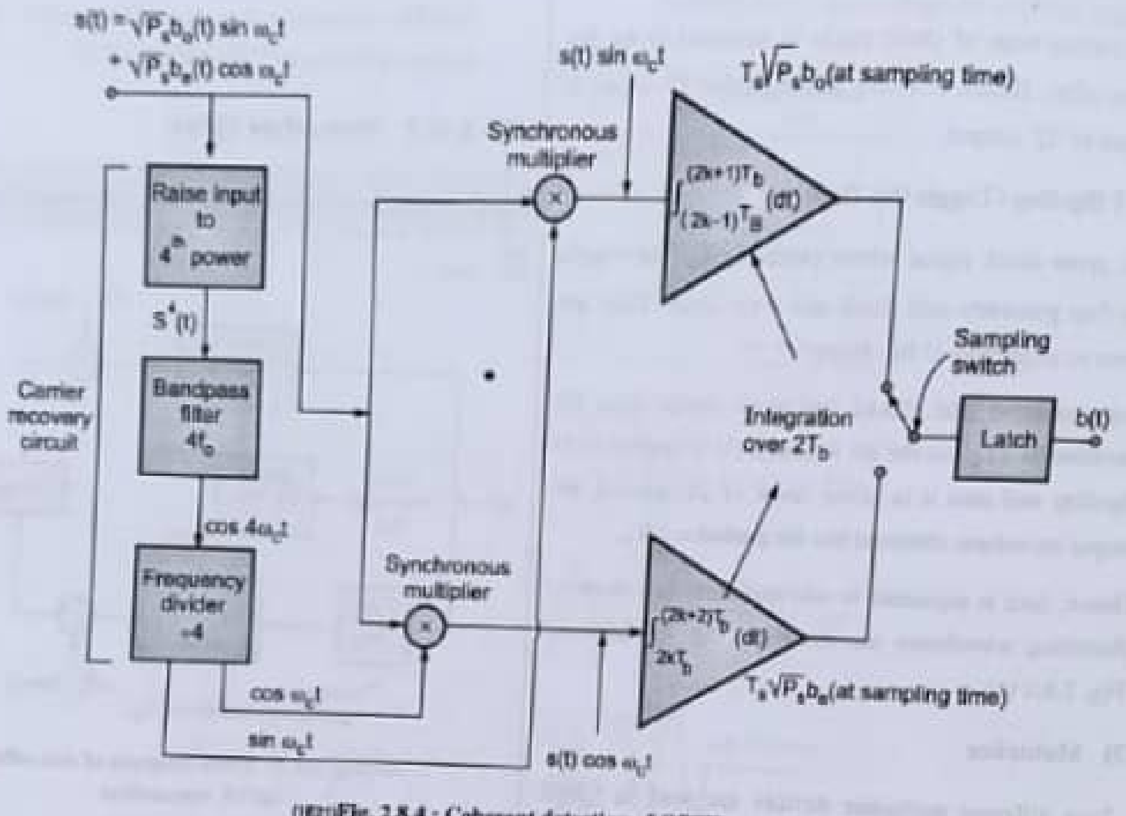
- The signals $S_o(t)$ and $S_e(t)$ will have the phase shifts at the same time.
- Due to this, amplitude variations in the output waveform occur at the same time.
- These amplitude variations are more prominent in non-offset QPSK as compared to OQPSK.
- Due to these significant amplitude changes, non-offset QPSK system faces difficulties in using repeaters.

2.8.3 QPSK Receiver/Detection

UQ. Explain with block diagram QPSK receiver. SPPU - (E&TC Sem. 5) Dec. 15, 4 Marks

Ans. :

- *Refer Fig. 2.8.4*. It shows the block diagram of coherent detection of QPSK.



(1c21)Fig. 2.8.4 : Coherent detection of QPSK

1. Carrier recovery section.	2. Data recovery section.

(I) Carrier recovery section

- As the coherent detection technique is used, carrier has to be recovered separately at the receiver. This locally recovered carrier is then fed to the synchronous demodulators or multipliers used in data recovery section.
- The carrier recovery section is similar to that of BPSK receiver. The only change is everywhere factor of '4' is utilized like, 'Raise to 4^{th} power, BPF with $4f_c$, and frequency divider (÷ 4).
- Similar to BPSK receiver's carrier recovery section equations, it can be shown here also that carriers are recovered. Recovered carriers are [$\cos\omega_c t$] and [$\sin\omega_c t$].

▶ **(2) Data recovery section**

(i) Multipliers (ii) Integrators (2-bit interval integration).

(i) Multipliers and integration over 2-bit intervals

1. Incoming QPSK signal and the recovered carriers are multiplied. At multiplier 1,

$$V_{m1}(t) = V_{QPSK}(t)\cdot \sin\omega_c t$$
$$= [b_e(t)\sqrt{P_s}\cdot\sin\omega_c t + b_o(t)\sqrt{P_s}\cdot\cos\omega_c t]\sin\omega_c t$$
$$= b_o(t)\sqrt{P_s}\sin^2\omega_c t + b_e(t)\sqrt{P_s}\cdot\sin\omega_c t\cdot\cos\omega_c t$$

2. This $V_{m1}(t)$ is applied to integrator 1.Hence the output of integrator 1 is given by,

$$\text{Integrator's output} = \int_{(2k-1)T_b}^{(2k+1)T_b} \{b_o(t)\sqrt{P_s}\cdot\sin^2\omega_c t + b_e(t)\sqrt{P_s}\cdot\sin\omega_c t\cdot\cos\omega_c t\}\,dt$$

$$= \int_{(2k-1)T_b}^{(2k+1)T_b} b_o(t)\sqrt{P_s}\sin^2\omega_c t\,dt + \int_{(2k-1)T_b}^{(2k+1)T_b} b_e(t)\sqrt{P_s}\cdot\sin\omega_c t\cdot\cos\omega_c t\,dt$$

3. Now $\sin^2\omega_c t = \frac{1}{2}[1-\cos 2\omega_c t]$ and $\sin\omega_c t\cdot\cos\omega_c t = \frac{1}{2}\sin 2\omega_c t$ substituting these values in integration term we get,

$$\text{Integrator 1 output} = \int_{(2k-1)T_b}^{(2k+1)T_b} b_o(t)\sqrt{P_s}\left[\frac{1}{2}(1-\cos 2\omega_c t)\right]dt + \int_{(2k-1)T_b}^{(2k+1)T_b} b_e(t)\sqrt{P_s}\cdot\left(\frac{1}{2}\sin 2\omega_c t\right)dt$$

$$= \frac{b_o(t)\sqrt{P_s}}{2}\left\{\int_{(2k+1)T_b}^{(2k+1)T_b} 1\cdot dt - \underbrace{\int_{(2k+1)T_b}^{(2k+1)T_b} \cos 2\omega_c t\,dt}\right\} + \frac{b_e(t)\sqrt{P_s}}{2}\underbrace{\int_{(2k-1)T_b}^{(2k+1)T_b} \sin 2\omega_c t\,dt}$$

Integrals of sinusoidal functions over whole numbers = 0

$$= b_o(t)\sqrt{P_s}\,\frac{1}{2}[t]_{(2k-1)T_b}^{(2k+1)T_b} = b_o(t)\sqrt{P_s}\,\frac{1}{2}[2k\,T_b + T_b - 2k\,T_b + T_b]$$
$$= b_o(t)\sqrt{P_s}\,\frac{1}{2}[2T_b]$$
$$= b_o(t)\sqrt{P_s}\cdot T_b$$

(ii) Integrators (2-bit interval integration)

1. Similarly at the output of integrator 2 we get, [$b_e(t)\sqrt{P_s}\cdot T_b$]. Hence at the output, $b_o(t)$ and $b_e(t)$ are obtained.
2. Outputs of both the integrators are taken alternately with the help of switch. These bits are then stored in latch for duration T_b.
3. At the output of the latch 'b (t)' is obtained. As explained in BPSK, bit synchronizers are also needed in QPSK for bit interval detection.

2.8.4 Signal Space Representation of QPSK

GQ. Give mathematical representation of QPSK signal.

SPPU - (E&TC-Sem. 5) Dec. 16, May 17, 4 Marks

Ans. :

As we know,

$$V_{QPSK}(t) = \sqrt{2P_s} \cdot \cos\left[\omega_c t + (2m+1)\frac{\pi}{4}\right]$$

$$m = 0, 1, 2, 3 \qquad \text{...(2.8.1)}$$

- This can be expressed using two orthonormal basis functions.

$$V_{QPSK}(t) = \left[\sqrt{P_s T}\cos(2m+1)\frac{\pi}{4}\right]$$

As we know,

$$\cos(A+B) = \cos A \cdot \cos B - \sin A \cdot \sin B$$

$$\therefore \cos\left[\omega_c t + (2m+1)\frac{\pi}{4}\right]$$

$$= \cos\omega_c t \cdot \cos\left[(2m+1)\frac{\pi}{4}\right] - \sin\omega_c t \cdot \sin\left[(2m+1)\frac{\pi}{4}\right]$$

- Substituting in Equation (2.8.1) i.e. in $V_{QPSK}(t)$ equation we get,

$$V_{QPSK}(t) = \sqrt{2P_s}$$

$$\left\{\cos\omega_c t \cdot \cos\left[(2m+1)\frac{\pi}{4}\right] - \sin\omega_c t \cdot \sin\left[(2m+1)\frac{\pi}{4}\right]\right\}$$

$$= \left\{\sqrt{P_s T}\cos\left[(2m+1)\frac{\pi}{4}\right]\sqrt{\frac{2}{T}}\cos\omega_c t\right\}$$

$$-\left\{\sqrt{P_s T}\sin\left[(2m+1)\frac{\pi}{4}\right]\sqrt{\frac{2}{T}}\sin\omega_c t\right\} \qquad \text{... (2.8.2)}$$

- Hence comparing with original QPSK output equation we get,

$$b_e(t) = \sqrt{2}\cos\left[(2m+1)\frac{\pi}{4}\right]$$

and

$$b_o(t) = -\sqrt{2}\sin\left[(2m+1)\frac{\pi}{4}\right]$$

- Expressing in terms of ortho normal functions,

$$\phi_1(t) = \sqrt{\frac{2}{T}}\cos\omega_c t$$

$$\text{and} \quad \phi_2(t) = \sqrt{\frac{2}{T}}\sin\omega_c t$$

$$\therefore V_{QPSK}(t) = \sqrt{P_s T} \times \frac{1}{\sqrt{2}} b_e(t) \times \phi_1(t) + \sqrt{P_s T} \times \frac{1}{\sqrt{2}} b_o(t) \times \phi_2(t)$$

- $\phi_1(t)$ and $\phi_2(t)$ are orthonormal basis functions. For QPSK, symbol interval $T = 2T_b$.

$$\therefore V_{QPSK}(t) = \sqrt{2P_s \cdot T_b} \times \frac{1}{\sqrt{2}} b_e(t) \times \phi_1(t)$$

$$+ \sqrt{2P_s \cdot T_b} \times \frac{1}{\sqrt{2}} b_o(t) \times \phi_2(t)$$

As we know, $P_s \cdot T_b = E_b$ = energy of bit interval.

$$\therefore V_{QPSK}(t) = \sqrt{E_b}\, b_e(t)\,\phi_1(t) + \sqrt{E_b} \cdot b_o(t) \times \phi_2(t)$$

- Plot of this equation is signal space representation for QPSK. For plotting the signal space diagram please refer Table 2.8.1.

Table 2.8.1 : Plot of signal space

Symbol	$b_e(t)$	$b_o(t)$	$V_{QPSK}(t) = E_b \cdot b_e(t)\,\phi_1(t) + E_b \cdot b_o(t)\,\phi_2(t)$	Phases
00	−1	−1	$V_{QPSK}(t) = -E_b\phi_1(t) - E_b\phi_2(t)$	$\frac{3\pi}{4}$
01	−1	+1	$V_{QPSK}(t) = -E_b\phi_1(t) + E_b\phi_2(t)$	$\frac{5\pi}{4}$
10	+1	−1	$V_{QPSK}(t) = E_b\phi_1(t) - E_b\phi_2(t)$	$\frac{\pi}{4}$
11	+1	+1	$V_{QPSK}(t) = E_b\phi_1(t) + E_b\phi_2(t)$	$\frac{7\pi}{4}$

2.8.5 Power Spectra of QPSK

GQ. Draw power spectra of QPSK.

Ans. :

(1E27)Fig. 2.8.5(a) : Power spectra of baseband QPSK signal

Fig. 2.8.5(b) : Power spectra of QPSK signal

1. *Refer Fig. 2.8.5.* It shows power spectrum of QPSK. Power spectral density for QPSK is the sum of individual PSDs of in phase and quadrature components.
2. Hence, $S_x(f) = 2E \sin c^2 (T_f)$

$$\therefore S_x(f) = 4E_b \sin c^2 (2T_b f)$$

2.8.6 Bandwidth of QPSK

GQ. What is the bandwidth of QPSK system?

Ans. :

Bandwidth of QPSK is one half of the BPSK

$$\therefore BW = \frac{2f_b}{2} = f_b$$

2.8.7 Error Probability for QPSK System

UQ. Write an expression for its error probability of QPSK system. **SPPU - (E&TC-Sem. 5) Dec. 15, 4 Marks**

UQ. Draw signal space diagram of QPSK signal. Write the expression of all message points in the diagram. **SPPU - (E&TC-Sem. 5) Dec. 16, May 17, 4 Marks**

Ans. :

- As we know,

$$V_{QPSK}(t) = A \cos\left(\omega_c t + (2m+1)\frac{\pi}{4}\right),$$

$$m = 0, 1, 2, 3 \quad \text{for} \quad 0 \le t \le T_s = 2T$$

- Refer the signal space diagram drawn in Fig. 2.8.6. Also the receiver system is drawn in Fig. 2.8.7. Receiver diagram shows two correlators. These two correlators are provided with the locally generated carriers.

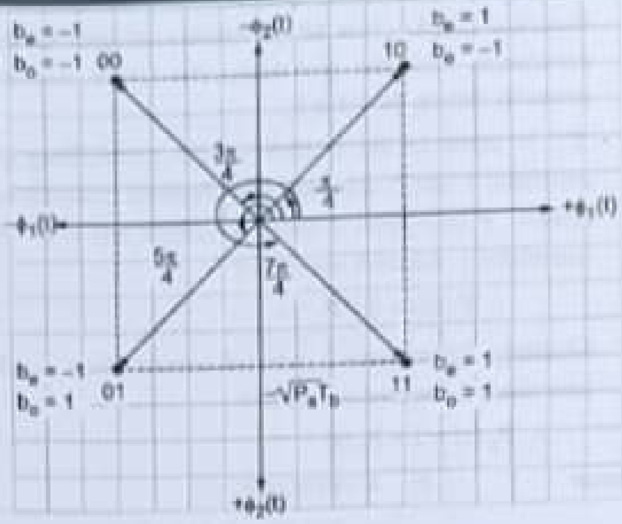


Fig. 2.8.6 : Signal space diagram for QPSK signal

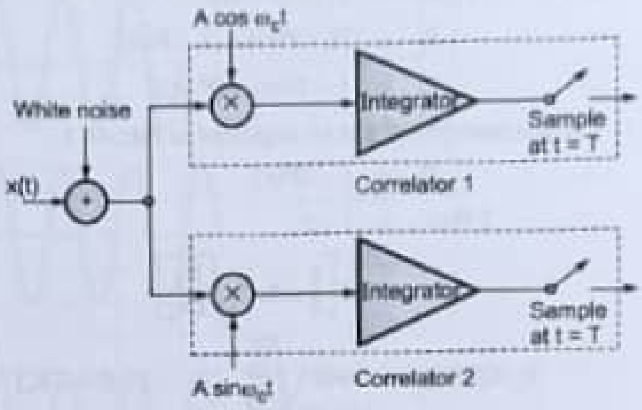


Fig. 2.8.7 : Block diagram of QPSK correlation receiver system

- From the receiver diagram, it is clear that the transmitted signal may be detected with the help of outputs of two correlators.
- There are chances that in presence of noise one of the correlator may give wrong output.
- Reference waveform of correlator 1 is at phase = 45° $= \frac{\pi}{4}$ as shown in Fig. 2.8.6.

Now probability that correlator 1 or 2 make an error is given by,

$$P_1(e) = P_2(e) = \frac{1}{2} \text{erfc} \sqrt{\frac{E_b}{N_o} \cos^2 \phi} \quad ...(2.8.3)$$

Substituting $\phi = 45°$, $\cos^2(45) = \frac{1}{2}$

$$\therefore P_1(e) = P_2(e) = \frac{1}{2} \text{erfc} \sqrt{\frac{E_b}{2N_o}} \quad ...(2.8.4)$$

- Equation (2.8.4) gives bit probability of error of the correlators. Now the probability that QPSK receiver correctly detects the bit is P_c.
- 'P_c' is equal to the product of the probabilities of correlator 1 and 2 such that they detected correct bits. Hence, P (correlator 1 detected correct bit) = 1 – P (correlator 1 detected wrong bit)

$$\therefore P_1(c) = 1 - P_1(e)$$

Similarly, P(correlator 2 detected correct bit) = 1 – P (correlator 2 detected wrong bit)

$$\therefore P_2(c) = 1 - P_2(e)$$

Hence $\therefore P_c = P_1(c) \times P_2(c)$

$$= [1 - P_1(e)] \times [1 - P_2(e)]$$

But $P_1(e) = P_2(e) = P(e)$

$$\therefore P_c = [1 - P(e)] \times [1 - P(e)]$$

$$\therefore P_c = 1 - 2P(e) + [P(e)]^2$$

- The last term $[P(e)]^2$ can be neglected as $P(e) << 1$.

$$\therefore P_c = 1 - 2P(e)$$

$$\therefore 2P(e) = 1 - P_c$$

$$= 2\left[\frac{1}{2}\operatorname{erfc}\sqrt{\frac{E_b}{2N_o}}\right]$$

$$\therefore P_e(QPSK) = \operatorname{erfc}\sqrt{\frac{E_b}{N_o}} \quad ...[\because E = 2E_b]$$

Euclidean distance from signal space diagram

- Euclidean distance measures the ability of the receiver to detect the bit without error. Hence, signal points which differ by single bit are separated by distance.

$$d = 2\sqrt{P_s \cdot T_b} = 2\sqrt{E_b}$$

- This distance is same as that of BPSK. Hence noise immunity of both systems is same.

2.8.8 Advantages, Disadvantages and Applications of QPSK

GQ. State advantages, disadvantages and applications of QPSK.

Ans. :

Advantages

(1) Better noise immunity.

(2) Effective utilization of available spectrum as 2bits/symbol. Also it is bandwidth efficient system.

(3) Error probability is low.

Disadvantages

(1) Generation and detection is complicated.

(2) It is not power efficient shift keying technique compared to other techniques.

Applications

(1) Various cellular wireless standards like CDMA, 802.11 WLAN, 802.16 fixed and mobile Wi Max.

(2) Satellite communication.

(3) Cable TV.

2.8.9 Comparison between Offset QPSK and Non-offset QPSK

GQ. Compare offset QPSK and non-offset QPSK.

Ans. :

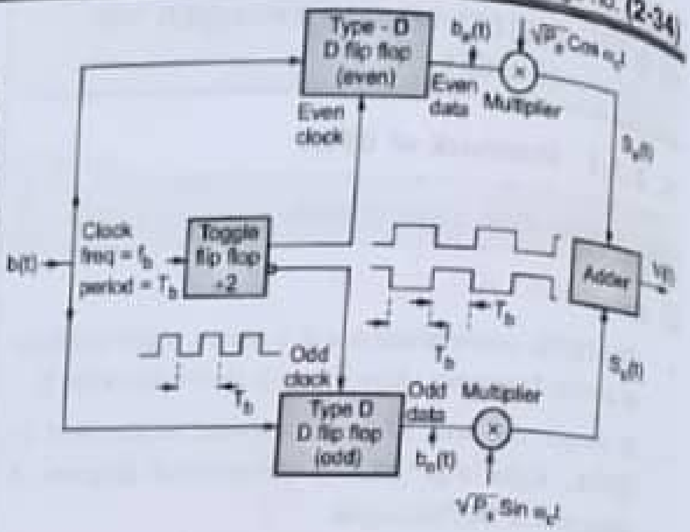
Sr. No.	Parameter	Non-offset QPSK	Offset QPSK
1.	Maximum phase change	± 180° for standard QPSK	± 90°
2.	Linear or non-linear amplifier requirement	Required	Non-linear amplifier needed.
3.	Amplitude variations at the instants of abrupt phase changes	Medium	Small
4.	Simultaneous phase change of I and Q phases	Yes	No.
5.	Offset between I and Q phases	No	Yes

2.8.1 QPSK Transmitter/Generation

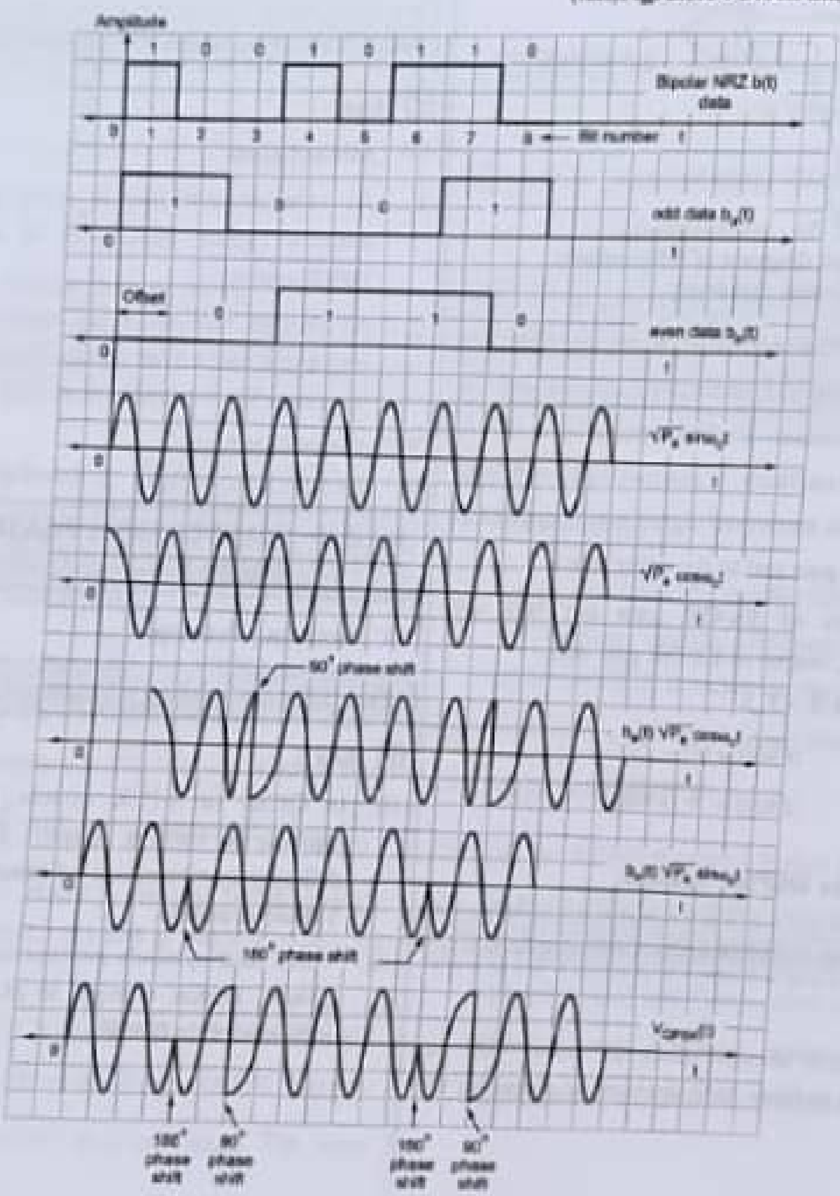
> **UQ.** Give mathematical representation of QPSK signal.
> **SPPU - (E&TC-Sem. 5) Dec. 16, May 17, 4 Marks**
>
> **UQ.** Explain QPSK signal generation.
> **SPPU - (E&TC-Sem. 5) Dec. 17, 6 Marks**

☑ **Ans. :**

- *Refer Fig. 2.8.1.* It shows block diagram of QPSK Transmitter *Refer Fig. 2.8.2.* It shows waveforms related to QPSK system.



(1E20)**Fig. 2.8.1 : Block diagram of QPSK transmitter**



(1E17)**Fig. 2.8.2 : Waveforms of offset QPSK**

2.9 M-ARY PSK MODULATION

Working principle

GQ. State working principle of M-ary PSK.

UQ. Draw and explain signal space representation of following signal : 8 ary PSK. [Hint : Take M = 8]

SPPU - (E&TC-Sem. 5) Dec. 15, 4 Marks

Ans. :

- It BPSK, symbol comprises of single binary bit. In QPSK, two bits are present per symbol. Also, in BPSK, phases are obtained by $\left[\frac{2\pi}{2} = 180°\right]$. Hence carriers differ by 180°.
- Similarly in QPSK,

$$\text{Phases are} = (2m+1)\frac{\pi}{4} \quad \text{...m} = 0, 1, 2, 3.$$

- M-ary PSK is extension to these PSK systems. In this, 'N' bits together form a symbol. Hence, $2^N = M$ symbols are possible. Phases differ by $\frac{2\pi}{M}$.

2.9.1 M-ary PSK Transmitter

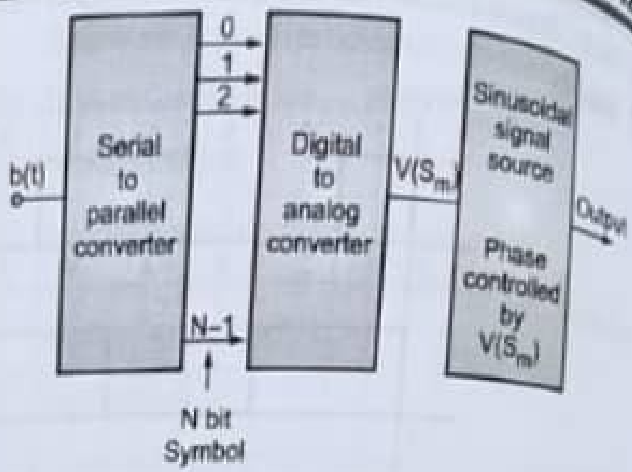
UQ. Explain block diagrams for generation of M-ary PSK signals.

SPPU - (E&TC-Sem. 5) Dec. 15, 2 Marks, Dec. 16, 6 Marks, May 17, 3 Marks

UQ. Explain M-ary PSK transmitter.

SPPU - (E&TC-Sem. 5) Dec. 17, 3 Marks

Ans. :



(1E15)Fig. 2.9.1 : M-PSK transmitter

Refer Fig. 2.9.1. It shows block diagram of M-PSK transmitter. It consists of :

(1) Serial-to-parallel convertor
(2) D/A convertor
(3) Sinusoidal signal source.

(1) Serial-to-parallel convertor

- The data bit stream **b (t)** is provided to it as input. This convertor can store 'N' bits. Hence, at the output N-bit symbol is obtained.
- Hence, convertor output remains unchanged for duration NT_b. Actually, during this NT_b duration convertor is assembling another N-bit symbol.

(2) D/A convertor

- N-bit symbol is applied to digital-to-analog convertor.
- It generates analog output voltage corresponding to one of $2^N = M$ different values. This is done for each symbol applied.

(3) Sinusoidal signal source

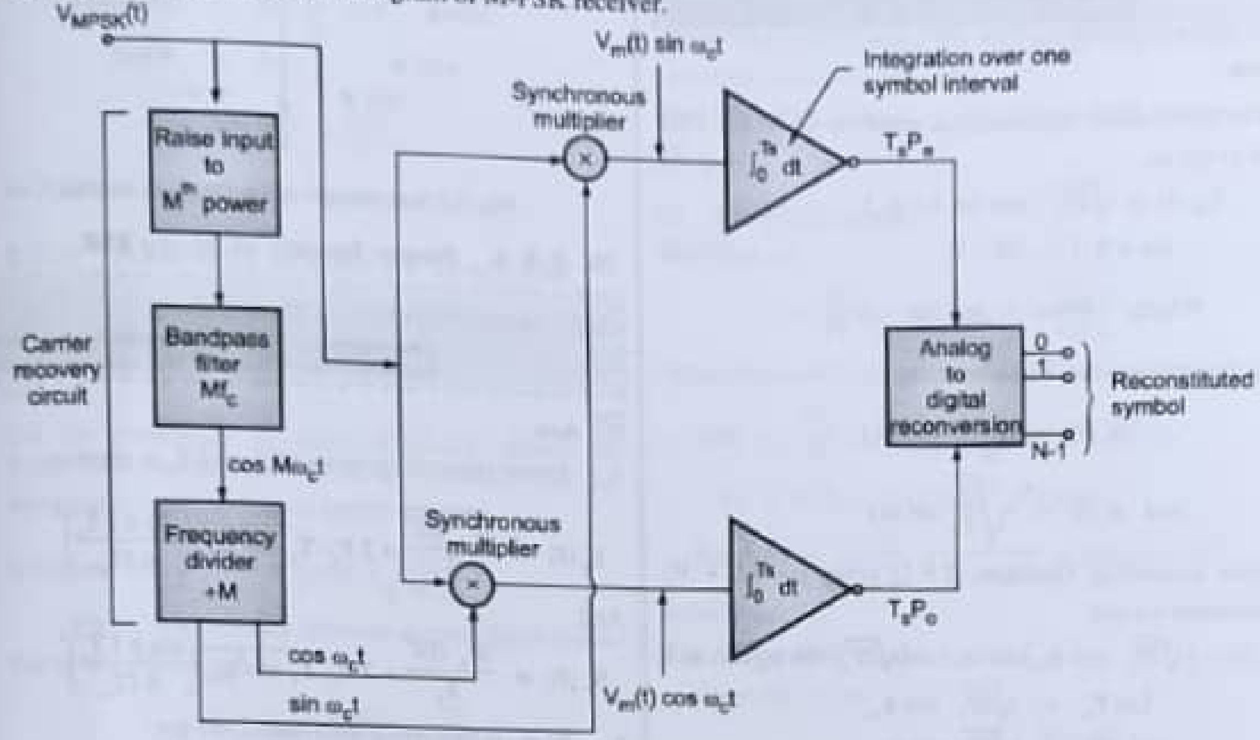
- The output of D/A convertor is applied to it.ϕ_m, i.e., phase of this sinusoidal source is controlled by the applied input voltage.
- Hence, the phase of this source has one-to-one correspondence with N-bit symbol. Thus at the output, 'M' different phases are available.

2.9.2 M-ary PSK Receiver

UQ. Explain block diagrams for reception of M-ary PSK signals.	SPPU - (E&TC-Sem. 5) Dec. 15, 2 Marks
UQ. Explain M-ary PSK receiver with suitable block diagram.	SPPU- (E&TC-Sem. 5) May 17, Dec. 17, 3 Marks

Ans. :

- *Refer Fig. 2.9.2.* It shows block diagram of M-PSK receiver.



(1E18)Fig. 2.9.2 : Block diagram of M-PSK receiver

This receiver design resembles to QPSK receiver. This is due to the fact that it is actually extension of BPSK and QPSK systems. Hence it has two sections:

(1) Carrier recovery section
(2) Data recovery section

(1) Carrier recovery section

- It comprises of :
  1. Raise to M^{th} power device
  2. BPF (Mf_c)
  3. Frequency divider (÷ M)
- Similar to QPSK receiver, received signal is applied to raise to M^{th} power device, filter and frequency divider.
- Frequency divider divides the incoming carrier frequency by a factor of 'M'. At the output, carriers are regenerated and provided to data recovery section.

(2) Data recovery section

It comprises of :

1. Multipliers
2. Integrators and bit synchronizers

3. A/D conversion

- Received signal and recovered carriers are multiplied at the multipliers. Then the signals are applied to integrators.
- Integrator integrates over symbol duration. For detection of start and end of symbol, bit synchronizer is needed.
- The output of integrator is the voltages. The amplitudes of these voltages are proportional to $T_s P_e$ and $T_s P_o$, respectively.
- Finally, amplitudes thus obtained are applied to A/D convertor. A/D convertor reconstructs the N bit symbol.

2.9.3 Geometric Representation of M-ary PSK/Signal Space Representation of M-ary PSK

> **UQ.** Explain suitable mathematical expressions and signal space representation of M-ary PSK signals.
> **SPPU - (E&TC-Sem. 5) Dec. 15, 2 Marks**

Ans. :

- The generalised mathematical equation of M-ary PSK is given as,

$$V_m(t) = \sqrt{2P_s} \cdot \cos(\omega_c t + \phi_m),$$
$$(m = 0, 1 \ldots, M-1) \qquad \ldots(2.9.1)$$

Where, Phase $= \phi_m (2m+1)\frac{\pi}{M}$

- Using orthonormal basis functions,

$$\phi_1(t) = \sqrt{\frac{2}{T_s}} \cos \omega_c t$$
and $$\phi_2(t) = \sqrt{\frac{2}{T_s}} \sin \omega_c t$$

- Now expanding Equation (2.9.1) using cos (A + B) formulae we get,

$$\therefore V_m(t) = (\sqrt{2P_s} \cdot \cos\phi_m)\cos\omega_c t - (\sqrt{2P_s} \cdot \sin\phi_m)\sin\omega_c t$$

Let $P_e = \sqrt{2P_s} \cdot \cos\phi_m$

and $P_o = \sqrt{2P_s} \cdot \sin\phi_m$

$$\therefore V_m(t) = P_e \cdot \cos\omega_c t - P_o \cdot \sin\omega_c t$$

- P_e and P_o change for every symbol $T_s = N \cdot T_b$. Thus, they have M values. *Refer Fig. 2.9.3* for signal space diagram of M-Ary PSK.

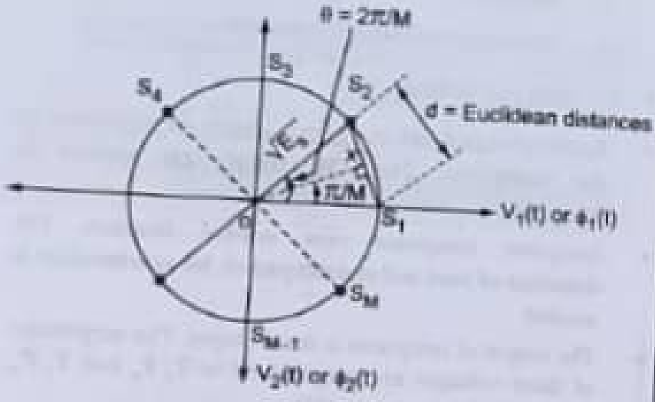


Fig. 2.9.3 : Signal space diagram M-Ary PSK receiver

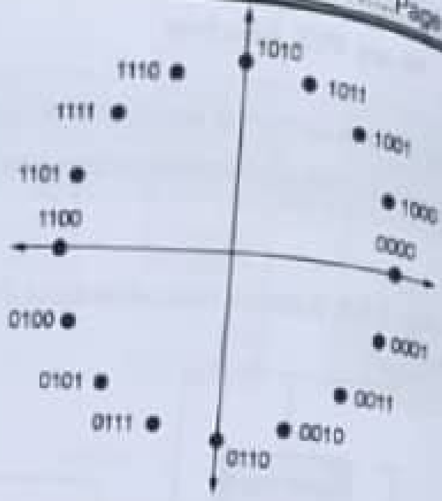


Fig. 2.9.3(a) : Single space diagram 16-PSK

2.9.4 Power Spectra of M-ary PSK

> **UQ.** Explain PSD of M-ary PSK signals.
> **SPPU - (E&TC-Sem. 5) Dec. 15, 2 Marks**

Ans. :

- Power spectral densities of P_e and P_o is given is

$$S_e(f) = \frac{\overline{|P_e(f)|^2}}{T_s} = 2P_s \cdot T_s \overline{\cos^2\phi_m}\left(\frac{\sin \pi f T_s}{\pi f T_s}\right)^2$$

And

$$S_o(f) = \frac{\overline{|P_o(f)|^2}}{T_s} = 2P_s \cdot T_s \overline{\sin^2\phi_m}\left(\frac{\sin \pi f T_s}{\pi f T_s}\right)^2$$

- As ϕ_m is uniformly distributed, we get,

$$\overline{\cos^2\phi_m} = \overline{\sin^2\phi_m} = \frac{1}{2}$$

$$\therefore S_e(f) = S_o(f) = P_s \cdot T_s\left(\frac{\sin \pi f T_s}{\pi f T_s}\right)^2$$

Refer Fig. 2.9.4 for power spectra

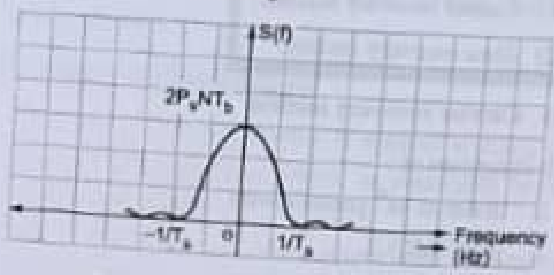


Fig. 2.9.4 : power spectra of M-Ary PSK

- Euclidean distance 'd',

$$d = \sqrt{4E_s \sin^2(\pi/M)} = \sqrt{4E_s \sin^2\left(\frac{\pi}{2^N}\right)}$$

- Hence as 'N' increases, distance between symbol decreases. Hence, probability of error increases.

2.9.5 Bandwidth of M-ary PSK

UQ. Explain Bandwidth of M-ary PSK signals.
SPPU - (E&TC-Sem. 5) Dec. 15, 2 Marks

Ans. :

$$\text{Bandwidth} = B = \frac{2}{T_s} = 2f_s$$

$$\text{But, } T_s = NT_b$$

$$\therefore B = \frac{2}{NT_b} = \frac{2f_b}{N}$$

As N increases, bandwidth decreases.

2.9.6 Probability of Error for M-ary PSK

GQ. Find probability error of 16-PSK.

Ans. :

- Let the probability of error of M-ary system be $P_e(M)$. The probabilities of M-ary systems can be represented by union bound approximation.
- It is given by, $P_e(M) \le \frac{1}{2} \text{erfc} \sqrt{\frac{d^2}{4N_o}}$

  Where, d = Distance between nearest signal points
- For M-ary PSK system,

$$d = d_{12} = d_{18} = 2\sqrt{E_s} \sin\frac{\pi}{M}$$

- Let symbol S_1 is transmitted, but due to error S_2 or S_8 are detected.

$$\therefore P_e(M) \le \frac{1}{2}\text{erfc}\sqrt{\frac{d_{12}^2}{4N_o}} + \frac{1}{2}\text{erfc}\sqrt{\frac{d_{18}^2}{4N_o}}$$

$$\le \text{erfc}\sqrt{\frac{d_{12}^2}{4N_o}} \quad \text{... [Since } d_{12} = d_{18}]$$

$$\le \text{erfc}\left[\frac{\left(2\sqrt{E_s}\cdot\sin\frac{\pi}{M}\right)^2}{4N_o}\right]^{1/2}$$

$$\le \text{erfc}\left[\frac{4E_s\sin^2\left(\frac{\pi}{M}\right)}{4N_o}\right]^{1/2}$$

$$P_e(M) \le \text{erfc}\left[\sqrt{\frac{E_s}{N_o}}\cdot\sin\left(\frac{\pi}{M}\right)\right]$$

Where, E_s = Symbol energy

- As we know, erfc (u) = $2Q\sqrt{2u}$

$$\therefore P_e(M) = 2Q\sqrt{2}\left[\sqrt{\frac{E_s}{N_o}}\cdot\sin\left(\frac{\pi}{M}\right)\right]$$

$$\therefore P_e(M) = 2Q\left[\sqrt{\frac{2E_s}{N_o}}\cdot\sin\left(\frac{\pi}{M}\right)\right]$$

This is expression of probability of error for M-ary PSK system.

Ex. 2.9.1 : In a digital communication system, the bit rate of a bipolar NRZ data sequence is 1 Mbps and carrier frequency of transmission is 100 MHz determine the symbol rate of transmission and the bandwidth requirement of the communications channel for:

(i) 8-ary PSK system. (ii) 16-ary PSK system.

Soln. :

[I] 8-ary PSK system

Here, M = 8

$$\therefore N = \log_2 M = \log_2 8 = \frac{\log 8}{\log 2} = 3$$

Hence there are 3 bits per symbol,

$$\therefore BW = \frac{2f_b}{N} = \frac{2\times1\times10^6}{3} = 6.66\times10^5 \text{ Hz}$$

$$\therefore T_s = NT_b = 3\times1\times10^{-6} = 3\ \mu\text{sec.}$$

$$\therefore \text{Symbol rate } f_s = \frac{1}{T_s} = \frac{1}{3\times10^{-6}} = 333.33\times10^{-3}$$

symbol/sec.

[II] 16-ary PSK system

Here, M = 16

$$\therefore N = \log_2 M = \log_2 16 = \frac{\log_{10}16}{\log_{10}2} = 4$$

Hence, there is 4 bit per symbol.

$$\therefore BW = \frac{2f_b}{N} = \frac{2\times1\times10^6}{N} = \textbf{500 kHz}$$

And symbol duration $Ts = NT_b = 4\times1\times10^{-6} = 4\ \mu\text{sec}$

$$\therefore \text{Symbol rate } f_s = \frac{1}{T_s} = \frac{1}{4\times10^{-6}}$$

$$= \textbf{250}\times\textbf{10}^3 \textbf{ symbol/sec.}$$

2.10 BINARY FREQUENCY SHIFT KEYING (BFSK)

GQ. State working principle of BFSK.

Working principle

- In BFSK system, binary bits '1' and '0' are represented by variations in frequencies. In this, only one bit is present per symbol.
- Practically, logic '1' is represented by 'f_H' and logic '0' by 'f_L' being lower frequency.

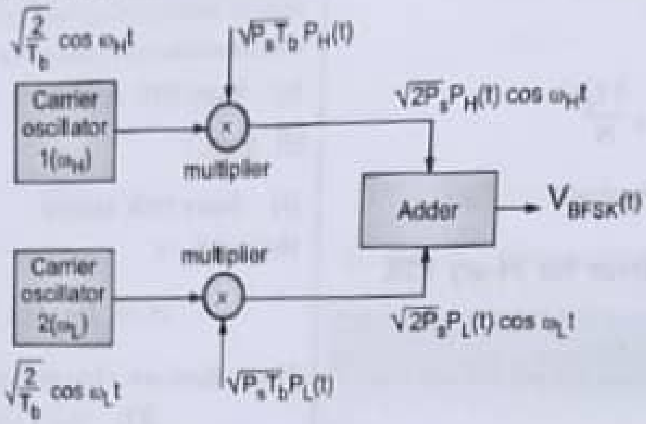
2.10.1 BFSK Transmitter

UQ. Explain coherent binary FSK signal generation. SPPU - (E&TC-Sem. 5) Dec. 17, 4 Marks

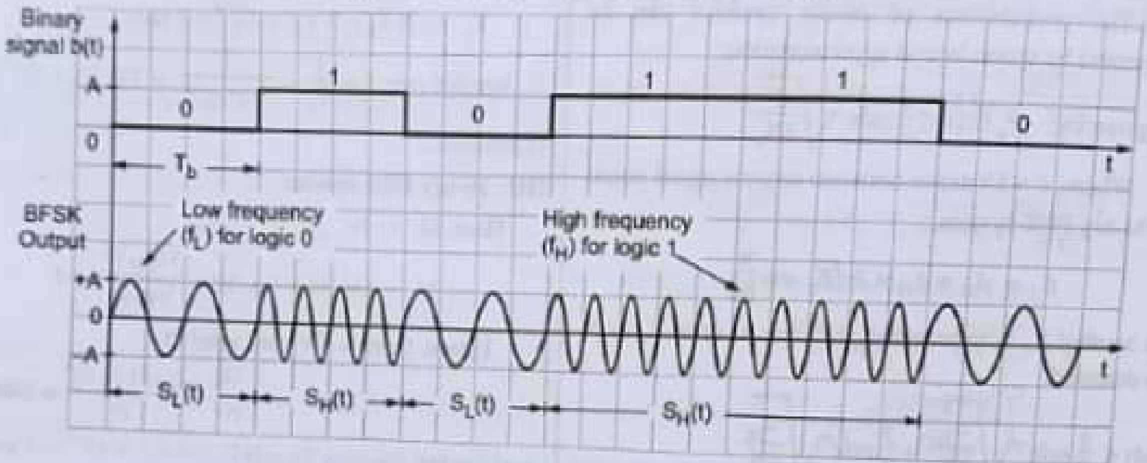
UQ. Explain generation of BFSK. SPPU - (E&TC-Sem. 5) May 18, 3 Marks

Ans. :

Refer Fig. 2.10.1. It shows block diagram of BFSK transmitter and Fig. 2.10.2 shows waveforms of BFSK.



(1E18)Fig. 2.10.1 : Block diagram of BFSK transmitter



(1E20)Fig. 2.10.2 : BFSK transmitter waveform

It consists of :

1. Carrier oscillators 2. Multipliers 3. Adder

1. Carrier oscillators

Two carrier oscillators are used to generate two different frequencies. The outputs of carrier oscillators are applied to the multipliers.

2. Multiplier

- Incoming carrier frequency and the applied bit are multiplied. Upper multiplier is provided with carrier ω_H and lower one with, ω_L.
- The voltage levels $P_H(t)$ and $P_L(t)$ are related to the d(t). It shows in Table 2.10.1.

Table 2.10.1 : Relationship between d(t) and $\frac{P_H(t)}{P_L(t)}$

d(t)	$P_H(t)$	$P_L(t)$
+1V	+1V	0V
–1V	0V	+1V

- When d(t) changes from +1 to – 1, $P_H(t)$ changes from 1V to 0V and $P_L(t)$ from 0V to 1V. Hence, at a time only one $P_H(t)$ or $P_L(t)$ is 1. Hence, the generated output is either ω_H or ω_L.

2.10.2 Mathematical Expression

GQ. State mathematical equation for BFSK.

Ans. :

$$V_{BFSK}(t) = \sqrt{2P_s}\cos[\omega_c t + d(t)\,\Omega t]$$

$$\therefore\ V_{BFSK}(t) = \sqrt{2P_s}\cos[\omega_c t + \Omega t]$$

... [d (t) = +1 for logic 1]

$$\text{And } V_{BFSK}(t) = \sqrt{2P_s}\cos[\omega_c t - \Omega t]$$

... [For d (t) = – 1 for logic 0]

$$\text{Let } \omega_H = \omega_c t + \Omega$$

$$\therefore\ V_{BFSK}(t) = \sqrt{2P_s}\cos\omega_H t$$

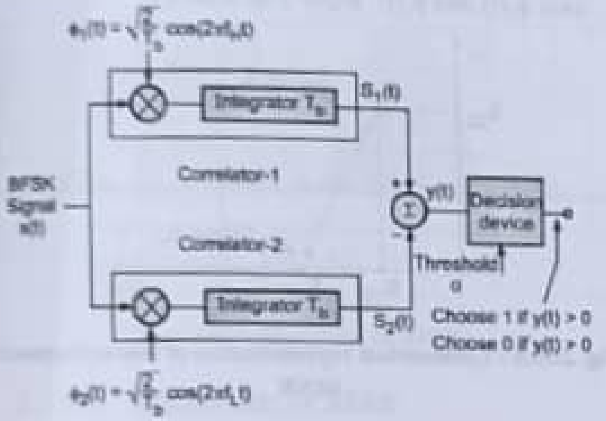
$$\text{And } V_{BFSK}(t) = \sqrt{2P_s}\cos\omega_L t$$

2.10.3 Coherent Detection of BFSK

UQ. Explain reception of BFSK.
SPPU - (E&TC-Sem. 5) May 18, 3 Marks

Ans. :

- *Refer Fig. 2.10.3.* **It shows coherent detection of BFSK.**



(1E10)**Fig. 2.10.3 : Coherent detection of BFSK**

The receiver consists of :

1. (Integrator and multiplier) Correlators
2. Adder
3. Decision device

1. Correlators

Two correlators are used for two different frequencies of FSK signal. Multiplier devices present in the correlators are provided with f_H and f_L frequency carriers.

2. Adder

The signals from integrators output are applied to the adder block. $S_1(t)$ and $S_2(t)$ are then added.

3. Decision device

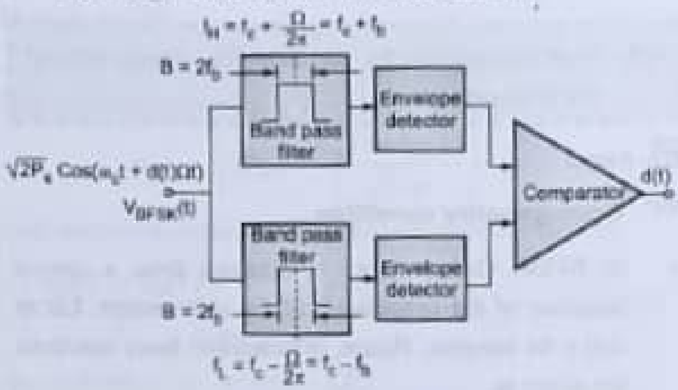
- If the transmitted frequency is f_H, then output $S_1(t)$ will be greater than $S_2(t)$.
- Hence, y (t) > 0 and bit 1 is detected and vice-versa.

2.10.4 BFSK Receiver (Non coherent BFSK) Receiver

UQ. Explain non-coherent Binary FSK.
SPPU - (E&TC-Sem. 5) May 18, 6 Marks

Ans. :

- *Refer Fig. 2.10.4.* It shows BFSK Receiver.



(1E11)**Fig. 2.10.4 : Block diagram of BFSK receiver using noncoherent detection**

- It consists of :

1. Two Band pass filters
2. Two Envelope detectors
3. Comparator

1. Band pass filters

- Upper BPF has centre frequency at f_H and the lower BPF have centre frequency at f_L. In BFSK receiver it is assumed that,

$$f_H - f_L = 2\left(\frac{\Omega}{2\pi}\right) = 2f_b$$

- The filter frequency ranges are selected such that they do not overlap. Each of the filters is able to accommodate the main lobe of the spectrum. Hence, one filter will pass all the signals of f_H and other will pass f_L.

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2. Envelope detectors

Filter outputs are applied to the envelope detector. As the name implies, it will detect the amplitude levels corresponding to incoming applied signal.

3. Comparator

- Envelope detector's outputs are applied to the comparator. Hence, both the inputs of comparator are compared. The bit corresponding to larger input will be detected as output.
- This means, if input 1 is high, **bit '1'** is detected. This is because it corresponds to f_H and vice-versa. Thus, d (t) is reconstructed. Practically, integrator and bit synchronizer is used.

2.10.5 Geometrical Representation of Orthogonal BFSK

GQ. State and explain the condition for orthogonality of the BFSK signal.

Ans. :

Orthogonality condition

- In BFSK, Orthogonality is obtained from a special selection of the frequencies of the unit vectors. Let m and n be integers. Hence, orthonormal basis functions are given as,

$$\phi_1(t) = \sqrt{\frac{2}{T_b}} \cos 2\pi m f_b\, t$$

$$\text{And} \quad \phi_2(t) = \sqrt{\frac{2}{T_b}} \cos 2\pi n f_b\, t$$

- Hence, vectors $\phi_1(t)$ and $\phi_2(t)$ are the m^{th} and n^{th} harmonics of the fundamental frequency f_b and $f_b = \frac{1}{T_b}$

Fourier analysis shows that, different harmonics $(m \pm n)$ are orthogonal over T_b.

- Let in BFSK, f_H and f_L are selected.

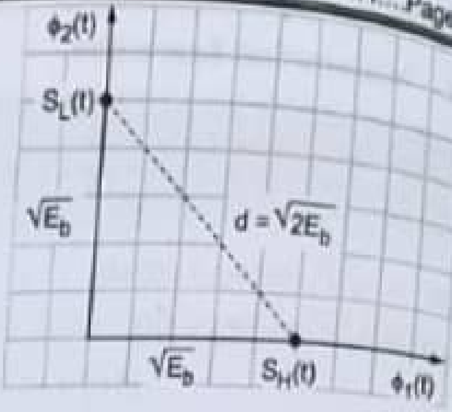
Assuming $m > n$

$$\therefore \; f_H = m f_b, \; f_L = n f_b$$

Hence signal vectors,

$$S_H(t) = \sqrt{E_b}\,\phi_1(t), \quad S_L(t) = \sqrt{E_b}\,\phi_2(t)$$

Signal space represented is plotted in Fig. 2.10.5.



(1E8)Fig. 2.10.5 : Signal space representation of BFSK

Euclidean distance between signal points is

$$d = \sqrt{2E_b}$$

2.10.6 Geometrical Representation of Non-orthogonal BFSK

GQ. Draw geometrical representation for non-orthogonal BFSK.

Ans. :

- When the two FSK signals $S_H(t)$ and $S_L(t)$ are not orthogonal, then the signal points will not lie exactly on axes $\phi_1(t)$ and $\phi_2(t)$. *Refer Fig. 2.10.6.*

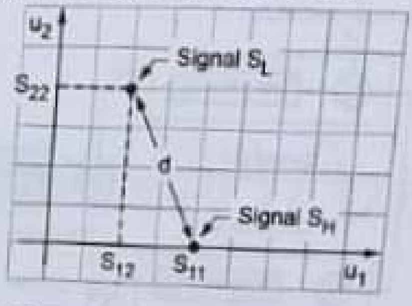


Fig. 2.10.6 : Geometrical representation of non-orthogonal BFSK

- The Euclidean distance 'd' is given by

$$d^2 \cong 2E_b\left[1 - \frac{\sin(\omega_H - \omega_L)\,T_b}{(\omega_H - \omega_L)\,T_b}\right]$$

2.10.7 Power Spectra of BFSK

GQ. Draw power spectra for BFSK modulated signal.

Ans. :

The output of BFSK is,

$$V_{BFSK}(t) = \sqrt{2P_s}\,P_H \cos(\omega_H t + \theta_H) + \sqrt{2P_s}\,P_L \cos(\omega_L t + \theta_L)$$

- Each of the two signals is assumed to be independent and uniformly distributed in phase is assumed. In BFSK, P_H and P_L are unipolar. Hence, it changes values between + 1 and 0.

$$\therefore P_H(t) = \frac{1}{2}+\frac{1}{2}P'_H(t); \quad P_L(t) = \frac{1}{2}+\frac{1}{2}P'_L(t)$$

- $P'_H(t)$ and $P'_L(t)$ are bipolar, so alternate values between + 1 and – 1.

$$\therefore V_{BFSK}(t) = \sqrt{\frac{P_s}{2}}\cos(\omega_H t+\theta_H)+\sqrt{\frac{P_s}{2}}\cos(\omega_L t+\theta_L)$$
$$+\sqrt{\frac{P_s}{2}}P'_H\cos(\omega_H t+\theta_H)+\sqrt{\frac{P_s}{2}}P'_L(\omega_L t+\theta_L)$$

- First two terms produce two impulses in their PSD; one at f_H and one at f_L. Last two terms produce spectrum of BPSK signals cantered at f_H and f_L. It is plotted in Fig. 2.10.7.

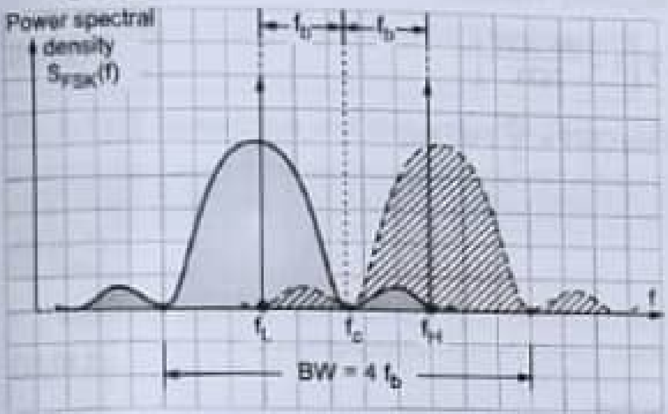


Fig. 2.10.7 : Spectrum of BFSK

2.10.8 Bandwidth of BFSK

GQ. Determine bandwidth requirement for transmission of signal.

☑ **Ans. :**

The bandwidth of BFSK signal is,

$$\boxed{B = 4f_b}$$

It is twice the bandwidth of BPSK.

2.10.9 Probability of Error of BFSK (Coherent Detection)

GQ. Derive equation for probability of error of BFSK for coherent detection.

☑ **Ans. :**

- Assume matched filter is used for the detection of BFSK signal. In FSK, the received signal is,

$$x_1(t) = A\cos(\omega_c+\Omega)t \quad \text{...(2.10.1)}$$
$$x_2(t) = A\cos(\omega_c-\Omega)t \quad \text{...(2.10.2)}$$

- As we know, correlation receiver acts as a matched filter when locally generated waveform is $[x_1(t)-x_2(t)]$

$$\therefore x_1(t)-x_2(t) = A\cos(\omega_c+\Omega)t - A\cos(\omega_c-\Omega)t \quad \text{...(2.10.3)}$$

Here, $A = \sqrt{2P_s}$

$$\therefore x_1(t)-x_2(t) = \sqrt{2P_s}\cos(\omega_c+\Omega)t - \sqrt{2P_s}\cos(\omega_c-\Omega)t$$

Now probability of matched filter is,

$$P_e = \frac{1}{2}\,\text{erfc}\left[\frac{x_{01}(T)-x_{02}(T)}{2\cdot\sqrt{2}\cdot\sigma}\right] \quad \text{... (2.10.4)}$$

And maximum signal to noise ratio for matched filter is,

$$\left[\frac{x_{01}(T)-x_{02}(T)}{\sigma}\right]^2_{max} = \frac{2}{N_0}\int_0^T x^2(T)\,dt \quad \text{...(2.10.5)}$$

As we know, $x(t) = x_1(t)-x_2(t)$

$$= \sqrt{2P_s}\,\{\cos(\omega_c+\Omega)t-\cos(\omega_c-\Omega)t\}$$

$$\therefore x^2(t) = \left[\sqrt{2P_s}\,\{\cos(\omega_c+\Omega)t-\cos(\omega_c-\Omega)t\}\right]^2$$
$$= 2P_s[\cos(\omega_c+\Omega)t-\cos(\omega_c-\Omega)t]^2$$

But $\cos(A+B)-\cos(A-B) = -2\sin A\sin B$

Using this trigonometric identity we get,

$$x^2(t) = 2P_s[-2\sin(\omega_c t)\sin(\Omega t)]^2$$
$$= 2P_s[4\sin^2(\omega_c t)\sin^2(\Omega t)]$$

Now, $2\sin^2\theta = 1-\cos 2\theta$

Using this identity we get,

$$x^2(t) = 2P_s[(1-\cos 2\omega_c t)(1-\cos 2\Omega t)]$$

Expanding the brackets we get,

$$x^2(t) = 2P_s[1-\cos 2\Omega t-\cos 2\omega_c t+\underbrace{\cos 2\omega_c t\cdot\cos 2\Omega t}_{\text{use } \cos A\cdot\cos B}]$$

We know, $\therefore \cos A \cos B = \frac{1}{2}[\cos(A-B)+\cos(A+B)]$

$$\therefore x^2(t) = 2P_s\left[1-\cos 2\Omega t-\cos 2\omega_c t+\frac{1}{2}\cos 2(\omega_c-\Omega)t+\frac{1}{2}\cos 2(\omega_c+\Omega)t\right] \quad ...(2.10.6)$$

Putting Equation (2.10.6) in Equation (2.10.5) we get,

$$\left[\frac{x_{01}(T)-x_{02}(T)}{\sigma}\right]^2_{max} = \frac{2}{N_o}\int_0^T x^2(t)\,dt$$

$$= \frac{2}{N_o}\int_0^T\left\{2P_s\left[1-\cos 2\Omega t-\cos 2\omega_c t+\frac{1}{2}\cos 2(\omega_c-\Omega)t+\frac{1}{2}\cos 2(\omega_c+\Omega)t\right]dt\right\} \quad ...(2.10.7)$$

$$= \frac{2}{N_o}\left\{\int_0^T 2P_s\,dt-\int_0^T 2P_s\cos 2\Omega t\,dt-\int_0^T 2P_s\cos 2\omega_c t\,dt+\frac{1}{2}2P_s\int_0^T\cos 2(\omega_c-\Omega)t\,dt+\frac{2P_s}{2}\int_0^T\cos 2(\omega_c+\Omega)t\cdot dt\right\}$$

$$= \frac{2}{N_o}\left\{2P_s T\left[1-\frac{\sin 2\Omega T}{2\Omega T}+\frac{1}{2}\frac{\sin[2(\omega_c+\Omega)T]}{2(\omega_c+\Omega)T}-\frac{1}{2}\frac{\sin[2(\omega_c-\Omega)T]}{2(\omega_c-\Omega)T}-\frac{\sin 2\omega_c T}{2\omega_c T}\right]\right\} \quad ...(2.10.8)$$

- If Ω (offset angular frequency) $<<\omega_c$, then last three terms will take form $\left[\frac{\sin 2\omega_c T}{2\omega_c T}\right]$ This ratio approaches zero as $\omega_c T$ increases. Assume $\omega_c T >> 1$. Hence, last three terms can be neglected. Therefore,

$$\left[\frac{x_{01}(T)-x_{02}(T)}{\sigma}\right]^2_{max} = \frac{2}{N_o}\times 2P_s T\left[1-\frac{\sin 2\Omega T}{2\Omega T}\right] \quad ...(2.10.9)$$

Maximum value of SNR is obtained when $2\Omega T=\frac{3\pi}{2}$. Hence, put $2\Omega T=\frac{3\pi}{2}$; we get,

$$\left[\frac{x_{01}(T)-x_{02}(T)}{\sigma}\right]^2_{max} = \frac{4P_s T}{N_o}\left[1-\frac{\sin\left(\frac{3\pi}{2}\right)}{\frac{3\pi}{2}}\right] \quad ...(2.10.10)$$

$$= (4.84)\frac{P_s T}{N_o} \quad ...(2.10.11)$$

Taking square root of both sides we get, $\left[\frac{x_{01}(T)-x_{02}(T)}{\sigma}\right]_{max} = \sqrt{\frac{4.84\,P_s\cdot t}{N_o}}$ ...(2.10.12)

Substituting in Equation (2.10.4) we get,

$$P_e = \frac{1}{2}\,\text{erfc}\left[\frac{x_{01}(T)-x_{02}(T)}{2\sqrt{2}\cdot\sigma}\right]=\frac{1}{2}\,\text{erfc}\left[\frac{1}{2\sqrt{2}}\sqrt{\frac{4.84\,P_s\cdot T}{N_o}}\right]=\frac{1}{2}\,\text{erfc}\left[\sqrt{\frac{0.6\,P_s\cdot T}{N_o}}\right]$$

$$P_e = \frac{1}{2}\,\text{erfc}\left[\sqrt{\frac{0.6E}{N_0}}\right] \quad ...[E=P_s\cdot T=\text{signal energy for bit interval}]$$

This is the expression for probability of error for **BFSK.**

Comments on results

If the signal energy in PSK signal is 0.6 times the signal energy of FSK signal, then equal error probabilities can be obtained in BPSK and BFSK systems.

GQ. Why FSK is less noise immune than PSK?

Ans. :

This is due to fact that, in PSK, $x_1(t) = -x_2(t)$. But this is not true in FSK. When optimum filter is used, PSK provides better results.

2.10.10 Probability of Error for Non Coherent Detection of FSK

GQ. State the probability of error for non-coherently detected BFSK.

Ans. :

The probability of error of non-coherent FSK is given as,

$$P_e = \frac{1}{2} e^{-\frac{E_b}{2N_o}}$$

2.10.11 Advantages, Disadvantages and Applications of BFSK

GQ. State advantages, disadvantages and applications for BFSK.

Ans. :

Advantages

1. Simple to generate
2. Compared to ASK, BFSK has better noise immunity.

Disadvantages

1. BFSK requires very high bandwidth compared to other shift keying techniques.
2. The bit error rate performance in presence of noise is worse compared to BPSK.

Applications

1. In FHSS system applications
2. Caller ID and remote metering applications.

2.11 M-ARY FSK

GQ. State working principle of M-ary FSK.

Ans. :

Working principle

It is the logical extension of BFSK. Actually BFSK is 2-FSK. This means M = 2. The number of bits per symbol is N and $2^N = M$. Hence at the output, 'M' different frequencies are obtained for representation of M different symbols.

2.11.1 Generation and Detection of M-ary FSK

GQ. Explain generation and detection of M-ary FSK.

Ans. :

- *Refer Fig. 2.11.1*. It shows M-ary FSK transmitter and receiver block diagram.

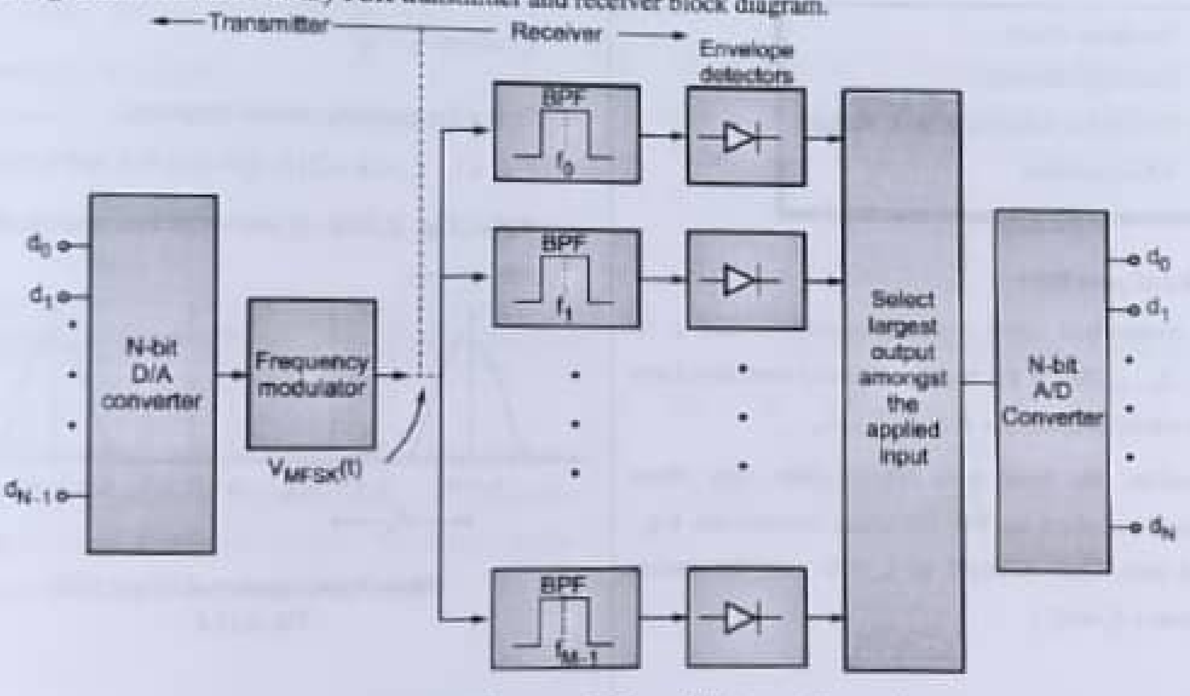


Fig. 2.11.1 : Block diagram of M-ary FSK transmitter receiver

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M-ary FSK transmitter

It consists of :

1. D/A converter
2. Frequency modulator

1. Digital to analog convertor (D/A converter)

N-bit symbol is applied at the input of the D/A converter. The duration of the symbol is T_s seconds. It converts N bit symbol into analog amplitude signal.

2. Frequency modulator

- The analog signal from convertor is applied to the frequency modulator. It acts as the modulating signal. Hence at the output of frequency modulator, frequency modulated carrier is obtained.
- The frequency of the carrier is changed in accordance with the input modulating signal. Hence, actually N-bit symbol controls the output frequency of a carrier.
- The transmitted signal for the symbol duration T_s is of the frequency

$$f_0 \text{ or } f_1 \text{ or } f_2 \text{ or } \dots f_{M-1}; \text{ where, } M = 2^N.$$

M-ary FSK Receiver : It consists of :

(i) Bandpass filters
(ii) Envelope detectors
(iii) Circuit for selecting largest output
(iv) A/D convertor

(i) Band pass filter

- The transmitted signal contains frequencies like f_0, f_1, f_2, ...f_{M-1}. Hence, the band pass filters used also have their centre frequencies at $f_0, f_1, f_2, \dots f_{M-1}$.
- Therefore, the band pass filters allow only those frequencies which are like the center frequencies. E.g., Band pass filter cantered at f_0 will pass the carrier frequency f_0 only.

(ii) Envelope detector

The output of Band pass filters is applied to the envelope detectors. Hence, the number of band pass filters used is equal to number of envelope detectors. As the name implies, it will detect the voltage signal.

(iii) Circuit for selecting largest output

The outputs of all the envelope detectors is applied to the device designed to select the largest output. The largest output amongst all the applied envelope detector's output is selected.

(iv) A/D converter

This is N bit A/D converter. The largest output selected is converted into digital bits. Hence, at the output N bit symbol is obtained.

2.11.2 Power Spectra of M-ary FSK

GQ. Explain M-Ary FSK with the help of following: (ii) Spectrum of M-Ary FSK.

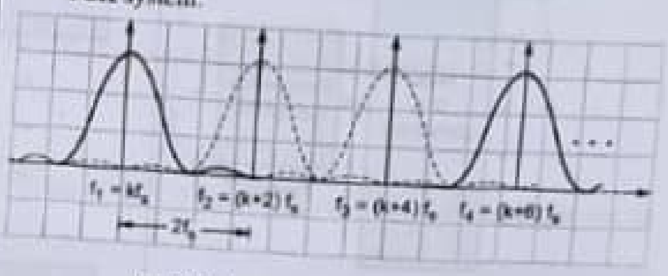
Ans. :

- To reduce the probability of error, frequencies f_0, f_1, f_{M-1} are selected such that M signals are orthogonal. For obtaining this, carrier frequency should be successive even harmonics of the symbol frequency f_s; where, $f_s = \frac{1}{T_s}$.

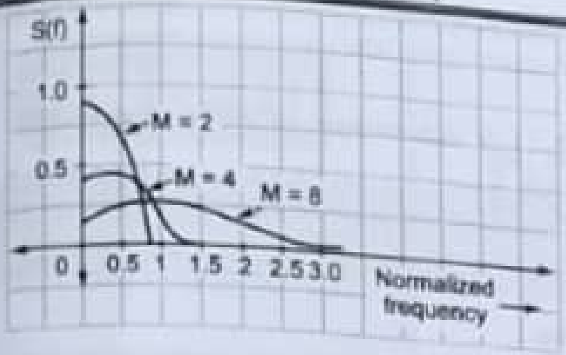
Hence for example, lowest frequency

$f_0 = k f_s$, $f_1 = (k+2) f_s$, $f_2 = (k+4) f_s$ and so on.

Refer Fig. 2.11.2. It shows power spectra of M-ary FSK system.



(1E8) (a) Power spectra of M-ary FSK
Fig. 2.11.2



(b) Power spectra of M-ary FSK for M = 2, 4 and 8
Fig. 2.11.2

- From Fig. 2.11.2. It is clear that the power spectra of individual possible transmitted signals overlap. It is extension of spectrum of BFSK.

2.11.3 Bandwidth of M-ary FSK

GQ. Explain M-Ary FSK with the help of following:
(iii) Bandwidth of M-Ary FSK.

Ans. :

Bandwidth of M-ary FSK is given as,

$$B = 2Mf_s$$

But, $f_s = \frac{1}{T_s}$ and $T_s = NT_b,\ T_b = \frac{1}{f_b}$

$$\therefore f_s = \frac{1}{NT_b} = \frac{f_b}{N}$$

$$\text{Also, } M = 2^N \quad \therefore B = 2^{N+1}\left(\frac{f_b}{N}\right)$$

Comments on result

Compared to M-ary PSK, the bandwidth of M-ary FSK is increased to a large extent.

2.11.4 Geometrical Representation of M-ary FSK

GQ. Draw signal space diagram for M-ary FSK.

Ans. :

The M-ary orthogonal FSK is extension of orthogonal BFSK. In this geometrical representation, 'M' mutually orthogonal co-ordinate axes are present. The signal vectors are then parallel to these axes. ***Refer Fig. 2.11.3.***

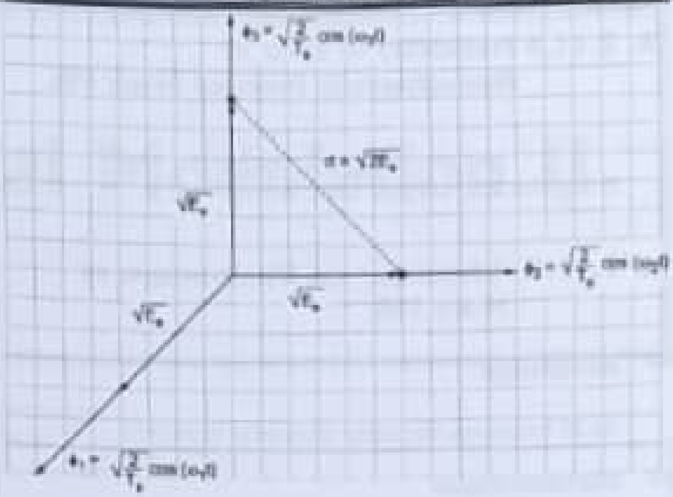


Fig. 2.11.3 : Geometrical representation of M-ary FSK

- Euclidean distance 'd' between signal points is given as,

$$d = \sqrt{2E_s}\ \sqrt{2NE_b} \quad ...\left[\begin{matrix} E_s = \text{symbol energy} \\ = NE_b \end{matrix}\right]$$

Here, 'd' for M-ary FSK is greater than 'd' for M-ary PSK except for M = 2 and M = 4.

2.11.5 Probability of Error for Coherently Detected M-ary FSK System

GQ. State probability of error for coherently detected M-ary FSK system.

Ans. :

- Signal space with 'M' dimensions is considered here. It is assumed that there is complete symmetry among signals. $S_1, S_2 ... S_M$. Hence probability of error is given as

$$P_e = (M-1)\frac{1}{2}\text{erfc}\sqrt{\frac{E_s}{2N_o}}$$

...[Extension of probability of error of BFSK]

- But, $M = 2^N$

$$\therefore P_e = \frac{2^N - 1}{2}\text{erfc}\sqrt{\frac{E_s}{2N_o}}$$

2.11.6 Probability of Error of Non-coherently Detected M-ary FSK

It is given by, $P_e \leq \frac{M-1}{2} e^{\left(-\frac{E_s}{2N_o}\right)}$

2.11.7 Advantages and Disadvantages of M-ary FSK

Advantage

As M increases, the noise immunity improves.

Disadvantages

1. It requires very large bandwidth. Also generation of carriers require large number of oscillators at the transmitter.
2. Precisely tuned BPF are needed at the receiver.

2.11.8 Advantages of MPSK over M-FSK

GQ. What are the advantages of M-ary PSK over M-ary FSK ?

Ans. :

1. M-ary PSK requires less bandwidth than M-ary FSK.
2. Precisely tuned BPF are not needed in case of M-ary PSK but they are important in M-ary FSK systems. It increases circuit complexity.

Chapter Ends...

□□□

3.1 AMPLITUDE SHIFT KEYING (BASK)

Working Principle

It is the simplest form of shift keying technique. Binary bits are multiplied with the carrier and two different amplitude levels are obtained at the output.

3.1.1 Generation and Detection of BASK

GQ. Explain BASK transmitter with neat block diagram.

Ans. :

(I) Block Diagram

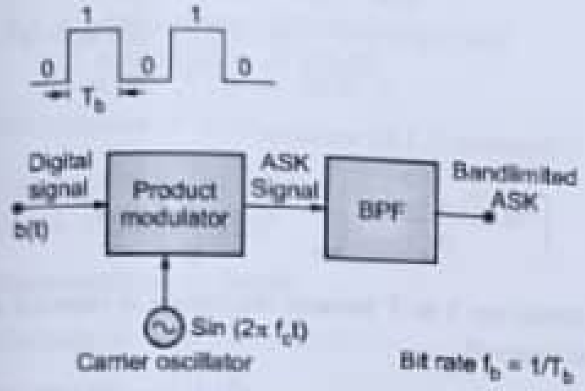


Fig. 3.1.1 : Block diagram of BASK transmitter

(i) *Refer Fig. 3.1.1.* It shows the block diagram of BASK transmitter. It consists of simple product modulator. **NRZ bipolar** signal is provided to the product modulator.

(ii) Sinusoidal carrier is also given to the product modulator. The output obtained after multiplication is BASK signal. It is shown in Fig. 3.1.2.

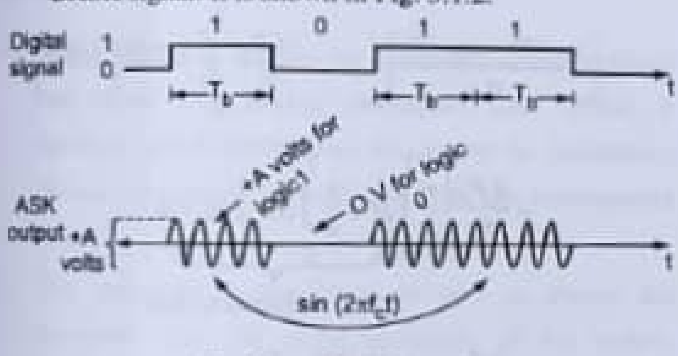


Fig. 3.1.2 : BASK waveform

(II) Mathematical expression

$$V_{ASK}(t) = d(t) \cdot \sin(\omega_c t)$$

$$\therefore V_{ASK}(t) = \sin(\omega_c t) \quad \text{...(for } d(t) = 1)$$

$$= 0 \quad \text{...[for } d(t) = 0]$$

3.1.2 BASK Receiver

GQ. Explain BASK receiver.

Ans. :

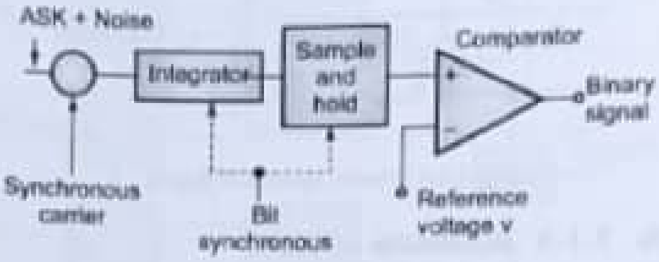


Fig. 3.1.3 : Block diagram of BASK receiver

(1) *Refer Fig. 3.1.3.* It shows the block diagram of BASK receiver. It is **coherent detection** technique.

(2) Synchronous carrier is multiplied with the received ASK signal. The product signal is then applied to the integrator.

(3) Integration is carried out over one bit interval. Then the output is applied to the **sample-and-hold** circuit. The sampled version of the integrator's output is then compared with the reference voltage 'V'.

(4) **If the S/H output < V → comparator's output is low. Hence binary bit '0' is detected and if S/H output > V → comparator's output is high; hence bit '1' is detected.**

3.1.3 Power Spectra of BASK

GQ. Draw the power spectrum of BASK.

Ans. :

(1) For obtaining power spectra of BASK, Fourier transform of BASK output is taken.

$$\therefore \mathbf{F}[V_{ASK}(t)] = \mathbf{F}[d(t) \sin(\omega_c t)]$$

$$= \frac{1}{2} \times (f + f_c) + \frac{1}{2} \times (f - f_c)$$

(2) Here **X (f) = sinc function = Fourier transform** of the rectangular input bits. Plot of power spectra is shown in Fig. 3.1.4.

Unit III End Sem

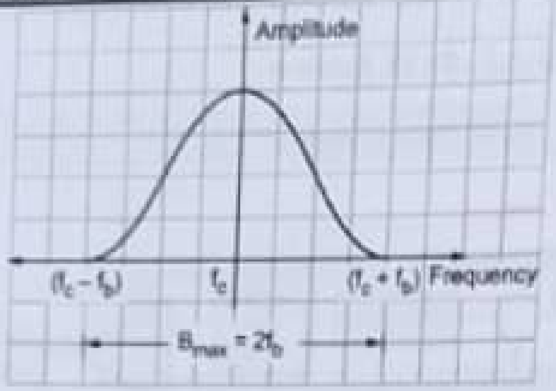


Fig. 3.1.4 : Power spectra of BASK

3.1.4 Bandwidth of BASK

GQ. State bandwidth of BASK.

Ans. :

(1) Bandwidth of BASK signal $= 2 f_b$ Hz

$$\therefore B = 2f_b$$

(2) The transmission bandwidth can be limited by the use of filter. This filter needs to be connected after ASK generator.

(3) Hence, the restricted **Bandwidth $= (1 + r) f_b$** Where r = Factor related to filter characteristics

3.1.5 Error Probability of BASK Signal

GQ. Derive the error probability of BASK.

Hint : To Derive $P_e = \frac{1}{2}$ erfc $\left[\sqrt{\frac{E_b}{4N_0}}\right]$

Ans. :

- In ASK, output is represented as,

Binary 1 : $x_1(t) = \sqrt{2P_s} \cdot \cos \omega_c t$

Binary 0 : $x_2(t) = 0$

- As we know, error probability of matched filter is,

$$P_e = \frac{1}{2} \operatorname{erfc}\left[\frac{x_{01}(T) - x_{02}(T)}{2 \cdot \sqrt{2} \cdot \sigma}\right] \quad ...(3.1.1)$$

- The expression for maximum SNR of optimum receiver is,

$$\left[\frac{x_{01}(T) - x_{02}(T)}{\sigma}\right]^2_{max} = \int_{-\infty}^{\infty} \frac{|X(f)|^2}{S_{ni}(f)} df \quad ...(3.1.2)$$

- Noise is assumed to be white. Hence, PSD of noise $S_{ni}(f) = \frac{N_0}{2}$ ...(3.1.3)

$$\therefore \left[\frac{x_{01}(T) - x_{02}(T)}{\sigma}\right]^2_{max} = \frac{2}{N_0} \int_{-\infty}^{\infty} |X(f)|^2 df \quad ...(3.1.4)$$

- By Rayleigh's theorem,

$$\int_{-\infty}^{\infty} |X(f)|^2 D_f = \int_{-\infty}^{\infty} x^2(t)\, dt \quad ...(3.1.5)$$

$$\therefore \left[\frac{x_{01}(T) - x_{02}(T)}{\sigma}\right]^2_{max} = \frac{2}{N_0} \int_{-\infty}^{\infty} x^2(t)\, dt$$

... [From Equations (3.1.5)] ...(3.1.6)

But $x(t) = x_1(t) - x_2(t)$

$= \sqrt{2P_s} \cdot \cos \omega_c t - 0$

$\therefore X(t) = \sqrt{2P_s} \cdot \cos \omega_c t$

$\therefore$ Equation (3.1.6) becomes,

$$\left[\frac{x_{01}(T) - x_{02}(T)}{\sigma}\right]^2_{max} = \frac{2}{N_0} \int_0^T x^2(t)\, dt$$

[limits are 0 to T because integration is expected over one bit interval]

$$\therefore \left[\frac{x_{01}(T) - x_{02}(T)}{\sigma}\right]^2_{max} = \frac{2}{N_0} \int_0^T [\sqrt{2P_s} \cdot \cos \omega_c t]\, dt$$

$$= \frac{2}{N_0} \int_0^T 2P_s \cos^2 \omega_c t\, dt = \frac{4P_s}{N_0} \int_0^T \cos^2 \omega_c t\, dt$$

Putting $\cos^2 \omega_c t = \frac{1 + \cos 2\omega_c t}{2}$

$$\left[\frac{x_{01}(T) - x_{02}(T)}{\sigma}\right]^2_{max} = \frac{4P_s}{N_0} \int_0^T \left[\frac{1 + \cos 2\omega_c t}{2}\right] dt$$

$$= \frac{4P_s}{N_0} \left\{\frac{1}{2}\int_0^T dt + \underbrace{\frac{1}{2}\int_0^T \cos 2\omega_c t\, dt}_{\text{Equal to zero}}\right\}$$

$$= \frac{2P_s}{N_0} [t]_0^T = \frac{2P_s \cdot T}{N_0}$$

$$\therefore \left[\frac{x_{01}(T) - x_{02}(T)}{\sigma}\right]_{max} = \sqrt{\frac{2P_sT}{N_0}}$$

But $P_s \cdot T = E_b$ = bit energy

$$\therefore \left[\frac{x_{01}(T) - x_{02}(T)}{6}\right]_{max} = \sqrt{\frac{2E_b}{N_0}}$$

Substituting in Equation (3.1.1) we get,

$$P_e = \frac{1}{2}\,\text{erfc}\left[\frac{1}{2\sqrt{2}}\sqrt{\frac{2E_b}{N_0}}\right]$$

$$P_e = \frac{1}{2}\,\text{erfc}\left[\sqrt{\frac{E_b}{4N_0}}\right]$$

This is the expression of bit error probability of ASK using matched filter.

3.1.6 Advantages, Disadvantage and Application of BASK

Advantages of BASK

1. Easy to generate and detect.
2. Simple to design system.

Disadvantage of BASK

The noise immunity is very low.

Application of BASK

It is used for low bit rate (< 100 bps) applications.

3.2 M-QAM/M-ARY QASK

(Quadrature Amplitude Modulation or Quadrature Amplitude Shift Keying)

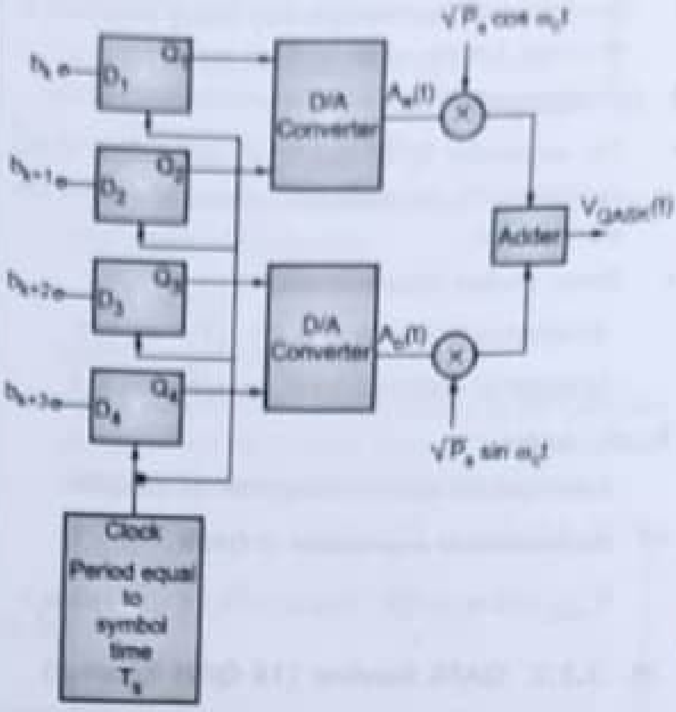
Working principle

(1) QAM system is actually the combination of amplitude and phase shift keying techniques. Like QPSK, it involves direct modulation of carriers in quadrature. Hence, advantages of ASK and PSK are incorporated in it.

(2) The variations in amplitude as well as phases are provided; thus, the noise immunity of the system increases.

3.2.1 16 - QAM Transmitter

(Generation and detection of M-ary QAM with M = 16)



(1E41)Fig. 3.2.1: Block diagram of QASK or 16 QAM transmitter

- *Refer Fig. 3.2.1.* It shows the block diagram of QASK or 16-QAM transmitter. The transmitter is designed to handle 4-bit symbol.

  Hence, **N = 4** therefore **$2^N = M = 2^4 = 16$ symbols** will be processed in it.

- It consists of :

(1) Register	(2) D/A convertor
(3) Multiplier	(4) Adder

(1) Register

- It is digital device made up of flip-flops. In this QAM transmitter, 4 D flip-flops are used as there are 4-bits per symbol.
- The 4-bit symbol $[b_{k+3}\, b_{k+2}\, b_{k+1}\, b_k]$ is stored in 4-bit register. The cascaded clock provided has duration equal to the symbol duration 'T_s'.
- Hence, new symbol is obtained once per interval $T_s = 4\,T_b$. The contents of the register are updated at each active clock edge.

Unit III End Sem

(2) D/A convertor

- The symbol formed with 4 bits with duration
- $T_s = 4\,T_b$ is applied to D/A convertors, as input.
- Convertor provides corresponding analog amplitude at the output. Let this output be $A_e(t)$ and $A_o(t)$.

(3) Multiplier

- The amplitudes $A_e(t)$ and $A_o(t)$ are applied to the multipliers. To the multipliers, sinusoidal carriers are also provided.
- Hence, product signal is obtained.

At output of Multiplier $= A_e(t) \cdot \sqrt{P_s} \cdot \cos \omega_c t$

At output of Multiplier 2 $= A_o(t) \cdot \sqrt{P_s} \cdot \sin \omega_c t$

(4) Adder

Adder adds the signals coming from the multiplier.

Mathematical expression of QASK

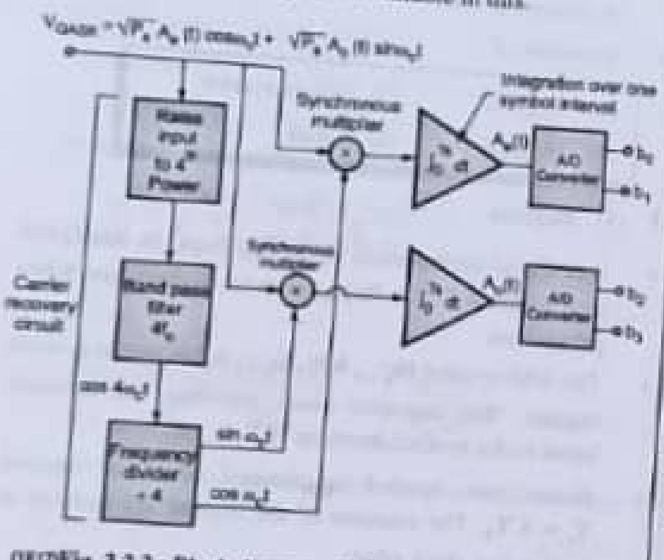
$$V_{QASK}(t) = A_e(t)\sqrt{P_s}\cdot\cos\omega_c t + A_o(t)\sqrt{P_s}\cdot\sin\omega_c t$$

3.2.2 QASK Receiver (16 QAM Receiver)

GQ. Explain QAM/QASK receiver in detail.

Ans. :

It is similar to QPSK receiver using coherent techniques. Similar to QPSK receiver, carrier recovery section recovers the carriers. These carriers are fed to the synchronous multipliers. But unlike QPSK receiver, amplitudes $A_e(t)$ and $A_o(t)$ are variable in this.



(1E17)Fig. 3.2.2 : Block diagram of QAM or QASK receiver

(1) Carrier recovery section

(i) Raise to 4th power output.

Received QASK signal is raised to 4th power.

$$\therefore \quad [V_{QASK}(t)]^4 = P_s^2\,\{A_e(t)\cos\omega_c t + A_o(t)\sin\omega_c t\}^4$$

(ii) At the output of BPF having cut off frequency $= 4 f_c$. We get,

All the components of $4 f_c$ are attenuated.

Hence,

$$\frac{V^4_{QASK}(t)}{P_s} = \left[\frac{A_e^4(t) + A_o^4(t) - 6A_e^2(t)A_o^2(t)}{8}\right]\cos 4\omega_c t + \left[\frac{A_e(t)A_o(t)\left[A_e^2(t) - A_o^2(t)\right]}{2}\right]\sin 4\omega_c t$$

(iii) From the above equation, it is clear that the average value of coefficient of $\cos 4\omega_c t$ is non-zero. However, the average value coefficient of $\sin 4\omega_c t$ is zero.

(iv) Hence, quadrature carriers $\sin\omega_c t$ and $\cos\omega_c t$ are obtained at the output of carrier recovery section.

(2) Data recovery section

(i) Recovered carriers are fed to respective synchronous demodulators or multipliers.

(ii) The integrator present integrates over one symbol duration. For this, symbol synchronizers are needed.

(iii) The output of integration will be $A_e(t)$ and $A_o(t)$. Hence, these analog signals are converted into digital ones by A/D convertors. Hence, at the output we get N-bit data stream.

3.2.3 Geometrical Representation of M-ary QAM Signal

UQ. Draw signal space of 16 - QAM system and comment on Euclidean distance for 16-QAM signals.
SPPU - (E&TC-Sem. 5) Dec. 16, 4 Marks
May 18, 2 Marks

UQ. Write the signal representation of M-QAM.
SPPU - (E&TC-Sem. 5) May 18, 2 Marks

☑ **Ans. :**

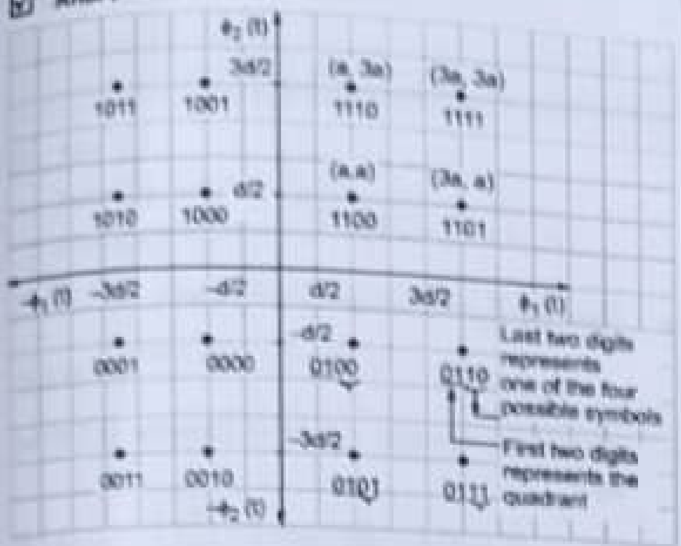


Fig. 3.2.3 : Signal space diagram of 16-QAM, (M=16)

(1) M-ary QAM can be viewed as two-dimensional generalization of M-ary PAM.

(2) It involves two **ortho- normal** basis functions.

$$\therefore \phi_1(t) = \sqrt{\frac{2}{T_s}}\cos\omega_c t \qquad 0 \le t \le T_s$$

$$\phi_2(t) = \sqrt{\frac{2}{T_s}}\sin\omega_c t \qquad 0 \le t \le T_s$$

(3) Now let d_{min} = **minimum distance** between 2 signal points.

a_k, b_k = integers = 1, 2, ...M

(4) Let $\sqrt{E_o} = \frac{d_{min}}{2}$ = energy of signal with lowest amplitude. Then the QAM output for k^{th} symbol is given by,

$$V_{QAM}(k\ T_s)\sqrt{\frac{2E_o}{T_s}}\ a_k \cos\omega_c t - \sqrt{\frac{2E_o}{T_s}}\ b_k \sin\omega_c t,$$
$$0 \le t \le T$$
$$k = 0, \pm 1, \pm 2 ...$$

(5) Equation shows two phase quadrature carriers. Each of the carriers will be modulated by a set of discrete amplitudes.

Euclidean distance, $d = \sqrt{0.4\ E_s}$

(6) **Comment :** The Euclidean distance is more in QAM as compared to PSK. The more the Euclidean distance, less is the error probability.

(7) Depending on number of possible symbols 'M', two distinct QAM constellations or signal space arrangements are available.

A.	QAM square constellations for M = even.
B.	QAM cross constellations for M = odd.

A. QAM square constellations

- It is applicable for even number of 'M'. Hence with even number of bits/symbol,

$$L = \sqrt{M}$$

where L = positive integer.

- Square constellation is always viewed as the Cartesian product of one-dimensional L-ary PAM constellations with itself. Hence square matrix can be written as,

$$(a_i, b_i) = \begin{bmatrix} (-L+1, L-1) & (-L+3, L-1) & \dots & (L-1, L-1) \\ (-L+1, L-3) & (-L+3, L-3) & \dots & (L-1, L-3) \\ \vdots & \vdots & & \vdots \\ (-L+1, -L+1) & (-L+3, -L+1) & \dots & (L-1, L+1) \end{bmatrix}$$

Example : 16 QAM system

Here M = 16, hence $L = \sqrt{M} = \sqrt{16} = 4$. Hence the Cartesian product of 4 PAM constellation will be,

$$(a_i, b_i) = \begin{bmatrix} (-3, 3) & (-1, 3) & (1, 3) & (3, 3) \\ (-3, 1) & (-1, 1) & (1, 1) & (3, 1) \\ (-3, -1) & (-1, -1) & (1, -1) & (3, -1) \\ (-3, -3) & (-1, -3) & (1, -3) & (3, -3) \end{bmatrix}$$

- As there are N = 4 bits per symbol, we have 16 combinations of bits using gray codes. The left most two bits → specify quadrant in (ϕ_1, ϕ_2) plane. Hence,

1^{st} quadrant	11
2^{nd} quadrant	10
3^{rd} quadrant	00
4^{th} quadrant	01

- The remaining two bits are used to represent one of four possible symbols lying within each quadrant of

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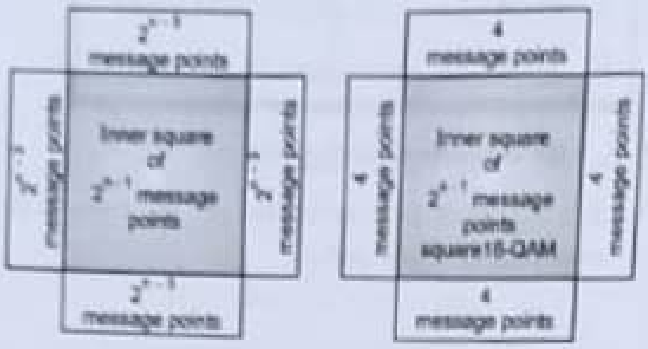
(ϕ_1, ϕ_2) plane. Accordingly signal space diagram plotted in Fig. 3.2.3.

B. QAM cross constellation



(a) Cross constellation in QAM **(b) Example of cross constellation - 16 QAM**

Fig. 3.2.4 : QAM cross constellation diagram

Procedures

It is applicable for odd numbers of bits per symbol. *Refer Fig. 3.2.4*

(1) Start with QAM square constellations with (n – 1) bits per symbol.

(2) Extend each side of QAM square constellations by adding 2^{n-3} symbols.

(3) Ignore the corners in the extension.

- The inner square shows 2^{n-1} symbols and four side extensions add $4 \times 2^{n-3} = 2^{n-1}$ symbols.
- Hence total number of symbols $= 2^{n-1} + 2^{n-1} = 2^n$

3.2.4 Probability of Error of M-ary QAM

UQ. Comment on probability of error for 16 - QAM signals. **SPPU - (E&TC-Sem. 5) Dec. 16, 4 Marks**

Ans. :

- The probability that receiver correctly detects the bit in M-ary QAM is,

$$P_c = (1 - P_e')^2 \quad ...(3.2.1)$$

where P_e' = probability of symbol error

- Hence, probability of symbol error is given as,

$$P_e' = \left(1 - \frac{1}{\sqrt{M}}\right) \text{erfc}\left(\sqrt{\frac{E_0}{N_0}}\right) \quad ...[L = \sqrt{M}] \quad ...(3.2.2)$$

- The probability of symbol error for M-ary QAM is given by,

$$P_e = 1 - P_c = 1 - (1 - P_e')^2$$

$$= 1 - (1 - 2P_e' + P_e'^2)$$

$$P_e = 1 - 1 + 2P_e' - P_e'^2$$

- Generally $P_e' \ll 1$, so ignoring $P_e'^2$ term,

$$\therefore \quad P_e \simeq 2P_e' \quad ...(3.2.3)$$

- Substituting Equation (3.2.2) in Equation (3.2.3) we get,

$$P_e = 2\left(1 - \frac{1}{\sqrt{M}}\right) \text{erfc}\left(\sqrt{\frac{E_0}{N_0}}\right) \quad ...(3.2.4)$$

- As amplitude variations are present in QAM system, transmitted energy is not constant. Hence, expressing 'P_e' in terms of average value of energy would be more appropriate rather than E_0.
- Let there are 'L' amplitude levels of the in phase and quadrature carriers.

$$\therefore \quad E_{av} = 2\left[\frac{2E_0}{L} \sum_{i=1}^{L/2} (2i-1)^2\right] \quad ...(3.2.5)$$

Expanding and simplifying above equation we get,

$$E_{av} = \frac{2(L^2 - 1) E_0}{3}$$

$$E_{av} = \frac{2(M-1) E_0}{3} \quad ...[\text{since } L = \sqrt{M}] \quad ...(3.2.6)$$

Substituting Equation (3.2.6) in Equation (3.2.4) we get,

$$P_e = 2\left(1 - \frac{1}{\sqrt{M}}\right) \text{erfc}\left(\sqrt{\frac{E_0}{N_0}}\right)$$

$$\text{But } E_0 = \frac{3E_{av}}{2(M-1)}$$

$$P_e = 2\left(1 - \frac{1}{\sqrt{M}}\right) \text{erfc}\left(\sqrt{\frac{3E_{av}}{2(M-1)N_0}}\right)$$

This is the expression for probability of error of M-ary QAM system.

3.2.5 Bandwidth of QAM

UQ. Find bandwidth requirement of M-QAM

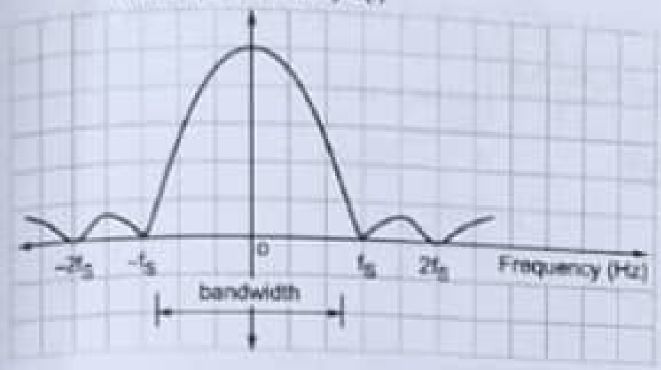
SPPU - (E&TC-Sem. 5) May 18, 2 Marks

Ans. :

Refer Fig. 3.2.5. It shows spectrum of QAM.

Bandwidth of QAM

$$B = f_s - (-f_s) = 2 f_s = \frac{2}{T_s} = \frac{2}{N T_b} = \frac{2f_b}{N}$$



(1E4)Fig. 3.2.5 : Frequency spectrum of QAM

3.3 MINIMUM SHIFT KEYING (MSK)

Generation and detection of MSK

3.3.1 MSK Transmitter

GQ. Explain the working of minimum shift keying modulator and demodulator, with the help of block diagram and waveform.

Ans. :

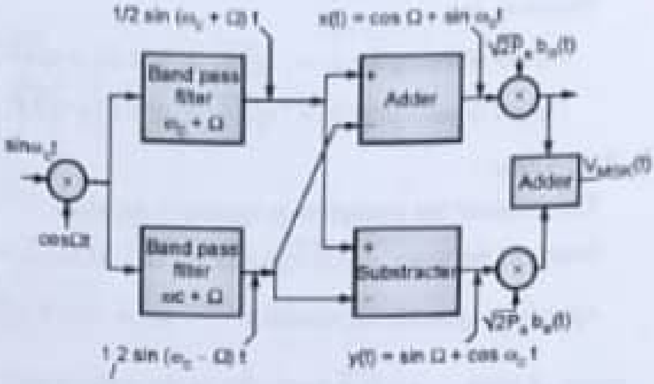
Working Principle

MSK is actually a form of **Continuous Phase FSK** (CPFSK). The basic concept of working is same as QPSK. Only the difference is instead of abrupt phase changes, MSK waveforms give **smooth** output. It also provides advantages like better **spectrum efficiency**.

Block Diagram of MSK

Refer Fig. 3.3.1. It shows block diagram of MSK transmitter. It consists of :

1. Product modulator (1)	2. Bandpass filters
3. Summation device	4. Product modulators (2)
5. Adder	



(1E2)Fig. 3.3.1 MSK transmitter

1. Product modulator (1)

- In this, carrier sin $\omega_c t$ and cos Ωt is multiplied. Hence, the two phase coherent sinusoidal waves at frequencies $(\omega_c + \Omega)$ and $(\omega_c - \Omega)$ are produced.
- Hence,

Output of multiplier = $\sin \omega_c t \cdot \cos \Omega t \sin \omega_c t \cdot \cos \Omega t = \frac{1}{2} [\sin(\omega_c + \Omega) t + \sin (\omega_c - \Omega) t]$

- It shows that multiplier output contains sum $(\omega_c + \Omega)$ and difference $(\omega_c - \Omega)$ frequency components.

2. Band pass filters

- Two band pass filters are used in MSK transmitter. Sum $(\omega_c + \Omega)$ is applied to BPF 1 and difference $(\omega_c - \Omega)$ and is applied to BPF 2.
- BPF1 is tuned to $(\omega_c + \Omega)$ and BPF 2 is tuned to $(\omega_c - \Omega)$. Therefore,

$$\text{Output of BPF1} = \frac{1}{2} \sin (\omega_c + \Omega)$$

$$\text{And Output of BPF2} = \frac{1}{2} \sin (\omega_c - \Omega)$$

3. Summation device

- Output of Band pass filters 1 and 2 is applied to the summation devices. These summation devices are used to linearly combine the filter outputs.
- At the output of summation devices, orthonormal basis functions x (t) and y (t) are obtained.
- These orthonormal basis functions are given as,

$$x(t) = \cos \Omega t \cdot \sin \omega_c t$$

$$\text{and } y(t) = \sin \Omega t \cdot \cos \omega_c t$$

4. Product modulators (2)

The signals x (t) and y (t) are multiplied with odd and even data bit streams, as shown in Fig. 3.3.1.

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Therefore, at

$$\text{Output of multiplier 2} = \sqrt{2P_s}\, b_o(t) \times x(t) = \sqrt{2P_s}\, b_o(t) \cos \Omega t \cdot \sin \omega_c t$$

$$\text{Output of multiplier 3} = \sqrt{2P_s}\, b_e(t)\, y(t) = \sqrt{2P_s}\, b_e(t) \sin \Omega t \cdot \cos \omega_c t$$

5. Adder

The output of the multipliers is applied to the adder.

$$\text{Output of the adder} = \left[\sqrt{2P_s}\, b_o(t) \cdot \cos \Omega t \cdot \sin \omega_c t\right] + \left[\sqrt{2P_s}\, b_e(t) \cdot \sin \Omega t \cdot \cos \omega_c t\right]$$

Applying trigonometric identities, $\sin A \cdot \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$

$$\therefore \text{output of adder} = \sqrt{2P_s}\, b_o(t) \cdot \frac{1}{2}[\sin(\omega_c + \Omega)t + \sin(\omega_c + \Omega)t] + \sqrt{2P_s}\, b_e(t) \cdot \frac{1}{2}[\sin(\omega_c + \Omega)t - \sin(\omega_c - \Omega)t]$$

$$= \sqrt{2P_s}\left[\frac{b_o(t) + b_e(t)}{2}\right] \sin(\omega_c - \Omega)t + \sqrt{2P_s}\left[\frac{b_o(t) - b_e(t)}{2}\right] \sin(\omega_c - \Omega)t$$

Let $C_H(t) = \frac{b_o(t) + b_e(t)}{2}$ and $C_L(t) = \frac{b_o(t) - b_e(t)}{2}$;

Also, $\omega_H = \omega_c + \Omega, \quad \omega_L = \omega_c - \Omega$

Hence, $V_{MSK}(t) = \sqrt{2P_s}\, C_H(t) \cdot \sin \omega_H t + \sqrt{2P_s}\, C_L(t) \sin \omega_L t$

This is the mathematical expression of MSK signal.

3.3.2 MSK Waveforms

GQ. Explain the working of minimum shift keying modulator and demodulator, with the help of block diagram and waveform.

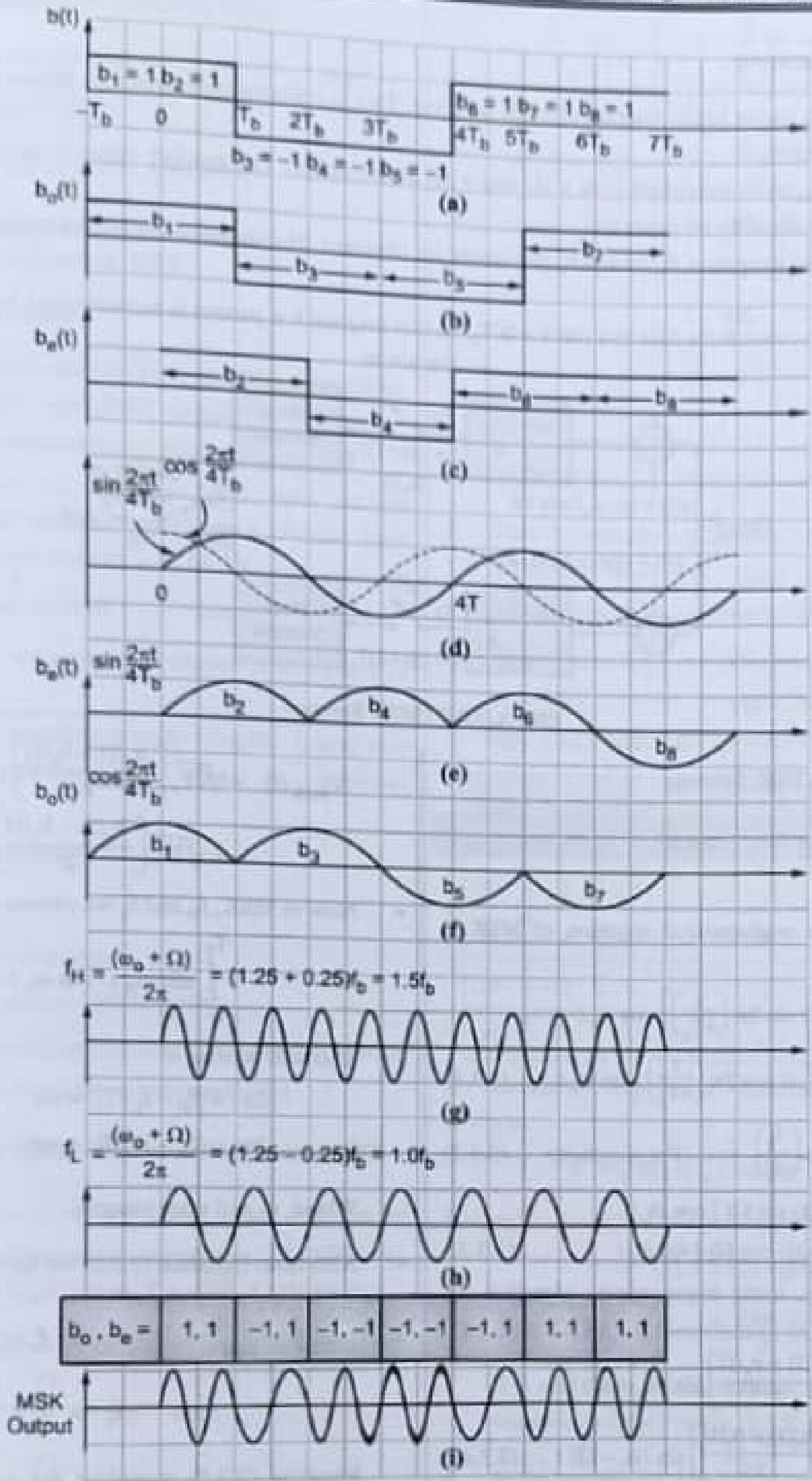
Ans. :

1. Refer Fig. 3.3.2. It shows MSK waveform. In Fig. 3.3.2(a). Bit stream **b (t)** with **$t = T_b$ = bit interval**
2. Fig. 3.3.2(b) Bit stream b (t) is divided into odd data bit stream b_o (t). Here, bit interval $T_s = 2T_b$
3. Fig. 3.3.2(c) – Bit stream b (t) is divided into even data bit stream b_e (t). Time interval of bit $T_s = 2\,T_b$ offset is observed in it.
4. Fig. 3.3.2(d) – The waveforms generated at MSK transmitter (i) $\sin 2\pi\left(\frac{t}{4T_b}\right)$ (ii) $\cos 2\pi\left(\frac{t}{4T_b}\right)$ These waveforms $\left[\sin 2\pi\left(\frac{t}{4T_b}\right)\right]$ crosses zero at the end of are (t) bit interval and $\left[\cos 2\pi\left(\frac{t}{4T_b}\right)\right]$ crosses zero at the end of b_o (t) bit interval.
5. Fig. 3.3.2(e) - It is the product of $b_e(t)$ and $\left[\sin 2\pi\left(\frac{t}{4T_b}\right)\right]$ waveforms. As the zero crossing is exactly at the end of bit interval, abrupt phase changes are not observed in the output.
6. Fig. 3.3.2(f) - It is the product of $b_o(t)$ and $\left[\cos 2\pi\left(\frac{t}{4T_b}\right)\right]$ waveforms. Similar to Fig. 3.3.3(e), this waveforms also does not show abrupt phase changes as in QPSK.
7. Fig. 3.3.2(g), (h) and (i) – These are f_H carrier, f_L carrier and $V_{MSK}(t)$ waveforms.

The MSK signal transmitted as,

$$V_{MSK}(t) = \sqrt{2P_s}\left[b_e(t) \sin 2\pi\left(\frac{t}{4T_b}\right)\right] \cos \omega_c t + \sqrt{2P_s}\left[b_o(t) \cos 2\pi\left(\frac{t}{4T_b}\right)\right] \sin \omega_c t$$

Hence in MSK, quadrature carrier $\cos \omega_c t$ and $\sin \omega_c t$ are multiplied with smoother waveforms of $\left[b_o(t) \cos 2\pi\left(\frac{t}{4T_b}\right)\right]$ and $\left[b_e(t) \sin 2\pi\left(\frac{t}{4T_b}\right)\right]$, respectively. Hence, there will be phase continuity. MSK waveform will not face abrupt phase changes.



(161)**Fig. 3.3.2 : MSK waveforms**

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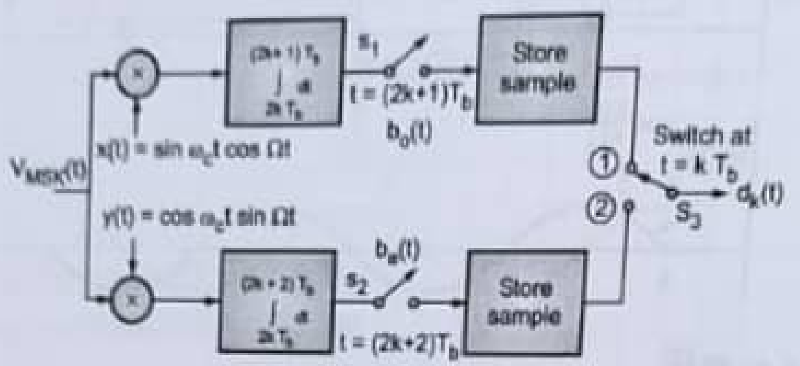
3.3.3 MSK Receiver

- Refer Fig. 3.3.3. It shows block diagram of MSK receiver. This is coherent type detection. Hence signals x (t) and y (t) are locally regenerated.
- The received $V_{MSK}(t)$ is multiplied with x (t) and y (t) at multiplier. The product signal is applied to the integrator. Integration occurs over two bit intervals.
- At the output of the integrator, $b_e(t)$ and $b_o(t)$ signals are obtained. The recovered $b_o(t)$ and $b_e(t)$ bits are sampled and stored.
- Switch S_3 is used to switch the data at a rate $t = KT_b$, so that original bit pattern is reconstructed back.



(1E3)Fig. 3.3.3 : MSK Receiver

3.3.4 MSK as FSK System

GQ. Why MSK is called FSK System?

Ans. :

- As we know, the mathematical equation of MSK is given as,

$$V_{MSK} = \sqrt{2P_s}\left[b_e(t)\sin 2\pi\left(\frac{t}{4T_b}\right)\right]\cos\omega_c t + \sqrt{2P_s}\left[b_o(t)\cos 2\pi\left(\frac{t}{4T_b}\right)\right]\sin\omega_c t \quad ...(3.3.1)$$

- Put $\Omega = \frac{2\pi}{4T_b} = 2\pi\left(\frac{f_b}{4}\right)$ ...For simplicity ... (3.3.2)

$$\therefore V_{MSK}(t) = \sqrt{2P_s}\,[\,b_e(t)\sin\Omega t\,]\cos\omega_c t + \sqrt{2P_s}\,[\,b_o(t)\cos\Omega t\,]\sin\omega_c t \quad ...(3.3.3)$$

- As shown already, using trigonometric identities we get,

$$V_{MSK} = \sqrt{2P_s}\left[\frac{b_o(t)+b_e(t)}{2}\right]\sin(\omega_c+\Omega)t + \sqrt{2P_s}\left[\frac{b_o(t)-b_e(t)}{2}\right]\sin(\omega_c-\Omega)t \quad ...(3.3.4)$$

- Putting $\omega_H = \omega_c + \Omega$,
  $\omega_L = \omega_c - \Omega$

$$V_{MSK}(t) = \sqrt{2P_s}\left[\frac{b_o(t)+b_e(t)}{2}\right]\sin\omega_H t + \sqrt{2P_s}\left[\frac{b_o(t)-b_e(t)}{2}\right]\sin\omega_L t \quad ...(3.3.5)$$

- Now in MSK, f_H and f_L are chosen such that,

$$\int_0^{T_b}\sin\omega_H t\cdot\sin\omega_L t\,dt = 0 \quad ...(3.3.6)$$

This will be true, if

$$2\pi = (f_H - f_L)\,T_b = n\pi$$

and

$$2\pi = (f_H + f_L)\,T_b = m\pi \quad ...(3.3.7)$$

Where, m and n are integers.

- Also from Equation (3.3.2) we have,

$$f_H = f_c + \frac{f_b}{4}$$

and

$$f_L = f_c - \frac{f_b}{4}$$

Equation (3.3.5) resembles the equation of FSK and hence, MSK appears as FSK system.

1. The above-mentioned properties of pulse shaping filter are achieved by passing NRZ binary data through a baseband filter.
2. The filter response should have Gaussian distribution. As a result, GMSK is obtained.
3. The transfer function of such a filter be,

$$h(t) = \sqrt{\frac{2\pi}{\log 2}}\, W\, e^{\left[-\frac{2\pi^2}{\log 2} W^2 t^2\right]}$$

where, W = 3dB baseband bandwidth of pulse shaping filter.

4. Therefore, impulse response can be given as,

$$H(f) = e^{\left[-\frac{\log 2}{2}\left(\frac{f}{W}\right)^2\right]}$$

... e.g., cell phones. It helps in reduction of battery consumption.

3. As GMSK is immune to amplitude variations, noise immunity of the system is better.

Disadvantages of Gaussian MSK

There are chances of intersymbol interference; but with BT = 0.3 parameter, it can be reduced. It requires complex channelization algorithms.

Applications of Gaussian MSK

GSM is TDMA based wireless standard. It makes use of **BT = 0.3GMSK**. 0.3 is used, as it provides best compromise between the increased bandwidth occupancy and resistance to co-channel interference.

3.5 COMPARISON OF DIGITAL MODULATION TECHNIQUES

GQ. Compare BPSK and QPSK.

Ans. :

Table 3.5.1 : Comparison between Digital Modulation Techniques

Sr. No.	Parameter	BASK	BPSK	DPSK	QPSK	M-ary PSK	QAM	BFSK	M-ary FSK
1	Variable characteristic in output carrier	Amplitude	Phase	Phase	Phase	Phase	Amplitude and phase	Frequency	Frequency
2	Mathematical expression	$=\sqrt{2P_s}\cos\omega_c t$ ...for 1 $=0$ ...for 0	$\pm b(t)\sqrt{2P_s}\cos(\omega_c t)$	$\pm\sqrt{2P_s}\cos(\omega_c t)$	$\sqrt{2P_s}\cos[\omega_c t + (2m+1)\frac{\pi}{4}]$ $M = 0,1,2,3$	$\sqrt{2P_s}\cos(\omega_c t + \phi_m)$ $\phi_m = (2m+1)\frac{\pi}{m}$	$K_1\sqrt{0.2P_s}\cos\omega_c t + k_2\sqrt{0.2P_s}\sin\omega_c t$	$\sqrt{2P_s}\cos(\omega_c t + \Omega t)\, d(t)$	$\sqrt{2P_s}\cos(\omega_i t)$ $i = 1,2,\ldots m$
3	Bits per symbols	1	1	1	2	N	N	1	N
4	Number of symbol	2	2	2	4	$M = 2^N$	$M = 2^N$	2	$M = 2^N$

Sr. No.	Parameter	BASK	BPSK	DPSK	QPSK	M-ary PSK	QAM	BFSK	M-ary FSK
5.	Bandwidth	$2f_b$	$2f_b$	f_b	f_b	$\frac{2f_b}{N}$	$\frac{2f_b}{N}$	$4f_b$	$\frac{2^{N+1}}{N} f_b$
6.	Error probability	$\frac{1}{2}\text{erfc}\sqrt{\frac{E_b}{4N_o}}$	$\frac{1}{2}\text{erfc}\sqrt{\frac{E_b}{N_o}}$	$\frac{1}{2}e^{\frac{-E_b}{N_o}}$	$\text{erfc}\sqrt{\frac{E_b}{N_o}}$	$\text{erfc}\left[\sqrt{\frac{E_s}{N_o}}\sin\frac{\pi}{M}\right]$	$\text{erfc}\sqrt{\frac{E_s}{N_o}}$	$\frac{1}{2}\text{erfc}\sqrt{\frac{0.6E}{N_o}}$	$\frac{M-1}{2}\text{erfc}\sqrt{\frac{E_s}{2N_o}}$
7.	Euclidean distanced	$\sqrt{E_b}$	$2\sqrt{E_b}$		$2\sqrt{E_b}$	$2\sqrt{E_s}\sin\frac{\pi}{M}$	$\sqrt{0.4E_s}$ for M = 16	$\sqrt{2E_b}$	$\sqrt{2NE_b}$
8.	Symbol duration (T_s)	T_b	T_b	$2T_b$	$2T_b$	NT_b	NT_b	T_b	NT_b
9.	Applications	low data rate applications	Telemetry		telephony, satellite communication	Satellite communication	Microwave digital radio, link, DVB	Paging services, cordless telephony	Paging services, cordless telephony

3.6 INTER SYMBOL INTERFERENCE (ISI)

3.6.1 Concept of ISI

UQ. What is Inter Symbol Interference (ISI)?

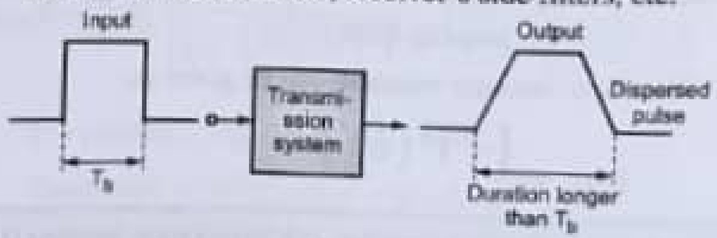
SPPU - (E&TC-Sem. 5) May 15(In Sem.), May 16(In Sem.), Dec. 16, 2 Marks

Ans. :

A communication channel is always band limited. Therefore, it always spreads or disperses the waveform passing through it.

3.6.1(A) Illustration of Concept of ISI

- Refer Fig. 3.6.1. It shows typical baseband digital system. There are many filters present in the system like transmitter's side filters, receiver's side filters, etc.



(1B44)Fig. 3.6.1 : Concept of ISI

- At the transmitter signal is